

# A computer-assisted proof candidate for the bound $\Lambda \leq 0.1787854$ for the de Bruijn–Newman constant\*

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## Abstract

We present a computer-assisted candidate for an unconditional proof of

$$\Lambda \leq \frac{893927}{5000000} = 0.1787854$$

for the de Bruijn–Newman constant. The argument instantiates the Polymath 15 barrier criterion at the exact parameters  $X = 6000000185827$ ,  $t_0 = 129/800$ , and  $y_0^2 = 87677/2500000$ . Its three inputs are the published Platt–Trudgian verification of the Riemann hypothesis to height 3000175332800, a finite-scan/analytic-tail nonvanishing argument for  $H_{t_0}$ , and a closed-prism interval-arithmetic barrier certificate. Seven computer-assisted propositions are tied to exact rational inputs, strict acceptance gates, and sealed evidence. Six computational lanes use FLINT/Arb. The load-bearing finite-error verifier and its promoted duplicate use the same pinned `mpmath.iv` 1.4.1 backend; because that interval module is documented as experimental, its transcendental enclosure semantics remain a release blocker pending audit or an Arb port. At repository commit [034a37acce703e128e63e2f36520b0c8af6c1c76](#), the repository records passing results for the stored quick verifier (including two promoted reconstruction programs), the canonical finite, tail, and barrier replay lanes, and a separate six-case mutation helper. A third promoted reconstruction source and its invocation schedule are sealed but not executed by the quick verifier. The exact theorem weld and all analytic reductions are set out here for review. Because independent expert validation of the full proof-to-code correspondence has not been completed, the displayed bound is a proof candidate rather than an established result. It would improve the peer-reviewed bound  $\Lambda \leq 0.2$  of Platt–Trudgian and the recent unrefereed preprint claim  $\Lambda \leq 0.1875$  of Mosaic Intelligence if validated; being positive, it would not prove the Riemann hypothesis.

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\*Version 1.1, 19 August 2026. The controlling computational record is the repository snapshot at commit [034a37ac](#). The argument is intended to be unconditional, but it has not completed independent expert review and is not presently an established theorem.

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# 1 Introduction

## 1.1 The de Bruijn–Newman constant

Let  $\Phi$  denote the super-exponentially decaying function

$$\Phi(u) = \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) \exp(-\pi n^2 e^{4u}),$$

and for  $t \in \mathbb{R}$  define the entire functions

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du. \quad (1)$$

The function  $H_0$  is a normalization of the Riemann  $\xi$ -function. With

$$\xi(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

analytic continuation of the Fourier identity gives, for every  $z \in \mathbb{C}$ ,  $H_0(z) = \frac{1}{8} \xi(\frac{1}{2} + \frac{iz}{2})$ , so that the Riemann hypothesis (RH) is the statement that all zeros of  $H_0$  are real. De Bruijn [1] proved that  $H_t$  has only real zeros for  $t \geq \frac{1}{2}$ . Newman [6] then proved that there is a finite constant  $\Lambda \leq \frac{1}{2}$ , the *de Bruijn–Newman constant*, such that  $H_t$  has only real zeros if and only if  $t \geq \Lambda$ . RH is equivalent to  $\Lambda \leq 0$ .

Newman conjectured  $\Lambda \geq 0$ , which was proved by Rodgers and Tao [9]. On the other side, de Bruijn’s classical upper bound  $\Lambda \leq 1/2$ , its strict improvement by Ki, Kim and Lee [4], and subsequent numerical work led to the bound

$$\Lambda \leq 0.22 \quad (2)$$

of the Polymath 15 project [8]. Subsequently Platt and Trudgian observed [7, Corollary 2] that their verification of RH to height  $3 \cdot 10^{12}$  (Section 3 below), combined with the second row of the parameter table in [8, Table 1, p. 65] (which requires verified height  $H > 2.51 \cdot 10^{12}$ ), yields the bound

$$\Lambda \leq 0.2, \quad (3)$$

which remains the relevant peer-reviewed benchmark for the present argument. Platt and Trudgian further note that the next entry of the same table would give  $\Lambda < 0.19$ , but only conditionally on a verified height slightly above  $10^{13}$ , beyond their theorem. The candidate value (5) lies below both 0.2 and 0.19 while consuming only the established  $3 \cdot 10^{12}$  verification height. A recent Zenodo preprint of Mosaic Intelligence [5] claims the intermediate value  $\Lambda \leq 0.1875$ ; it has not been peer reviewed. The value examined here is numerically smaller than both 0.2 and 0.1875. These comparisons do not promote either preprint claim to theorem status.

The complete review record, including the sealed candidate artifacts, reconstruction programs, invocation schedules, and mutation tests, is maintained in the access-controlled repository <https://github.com/judegomila/dbn-lambda-01787854-candidate-audit/tree/034a37acce703e128e63e2f36520b0c8af6c1c76>. All repository-specific assertions in this paper refer to commit `034a37acce703e128e63e2f36520b0c8af6c1c76` (tree `fe2d2f1149e9c24b7203d42305cccd642d386872`); later repository changes are not silently incorporated into this version. Every repository link in the paper is pinned to that immutable commit. A link identifies a location only; it does not imply inclusion in `SHA256SUMS`, execution by `verify.sh`, or independent status. Those distinctions are stated explicitly in Section 8.

## 1.2 Main result

The central claim examined in this paper is the following.

**Claim 1.1** (Computer-assisted candidate bound).

$$\Lambda \leq \frac{893927}{5000000} = 0.1787854.$$

The candidate proof instantiates the barrier criterion of [8, Theorem 1.2] at the exact parameters

$$X = 6000000185827, \quad t_0 = \frac{129}{800} = 0.16125, \quad y_0^2 = \frac{87677}{2500000} = 0.0350708, \quad (4)$$

with  $y_0 > 0$ , so that

$$t_0 + \frac{y_0^2}{2} = \frac{893927}{5000000} = 0.1787854. \quad (5)$$

One checks the elementary inequalities  $0 < t_0 < \frac{1}{2}$ ,  $0 < y_0^2 < 1 - 2t_0 = \frac{271}{400}$ , and  $y_0^2 + 2t_0 = \frac{893927}{2500000} < 1$ , which place the parameters in the domain of the criterion.

### 1.3 Structure of the proof, and the computer-assisted components

The candidate argument combines:

1. two published theorem inputs — the Polymath 15 criterion and effective approximation [8], and the Platt–Trudgian finite verification of RH [7];
2. a collection of new lemmas, with proofs given below, which bridge the published estimates to explicitly computable quantities; and
3. seven computer-assisted propositions tied to primary sealed evidence, totalling 3,149,013 finite “window” rows, a cell-cover computation, two uniform error bounds, a standalone tail computation, and an 883-prism barrier computation. Six lanes use FLINT/Arb ball arithmetic [3]; the finite-error lane instead uses the pinned `mpmath.iv` backend subject to the release-blocking caveat in Section 8.

The statements quoted from the published inputs in item (1) are Theorems 1.2 and 1.3 of [8] together with the displays (14), (20)–(24), (71)–(72), Proposition 6.6 and Lemma 8.4 of that paper, and Theorem 1 of [7]; all equation and theorem numbers below refer to the arXiv versions 1904.12438v2 and 2004.09765v1. The one point where our usage deliberately departs from a cited display, namely equation (24) of [8], is discussed in Remark 2.2. Each computer-assisted proposition of item (3) — the main finite computation (Proposition 4.3), the Dini cell computation (Proposition 4.9), the finite-region error bound (Proposition 4.10), the tail cutoff computation (Proposition 5.10), the barrier-box error bound (Proposition 6.2), the tail-exponent inequality (Proposition 6.5), and the barrier prism decomposition (Proposition 6.12) — is tied to primary sealed sources, certificates, logs, and verifiers in the candidate record. Each also has a derived standalone reconstruction in the referee workspace [dan-reworking/code/](#); three reconstruction sources have been promoted into the sealed [independent/](#) tree. These reconstructions are corroborative and are not the authority for the candidate assertions. The Proposition 4.10 program is only a same-backend refactoring of the corresponding primary routine, not an independent interval implementation. The quick stored verifier executes the Proposition 4.10 and 6.2 reconstructions; the much longer Proposition 4.3 reconstruction is governed by a recorded full-sweep schedule. The primary evidence, supplementary programs, and their authority levels are described with the respective propositions and summarized in Section 8.

## 2 The Polymath criterion and effective approximation

### 2.1 The barrier criterion

The logical skeleton of the proof is the following criterion.

**Theorem 2.1** (Polymath criterion, specialized from [8, Theorem 1.2]). *Let  $X > 0$ ,  $0 < t_0 < \frac{1}{2}$  and  $0 < y_0 < \sqrt{1 - 2t_0}$ . Suppose that:*

(i) (**Verified height.**) *There are no zeros  $\zeta(\sigma + iT) = 0$  with*

$$\frac{1 + y_0}{2} \leq \sigma \leq 1, \quad 0 \leq T \leq \frac{X}{2}.$$

(ii) (**Final-time right region.**) *There are no zeros  $H_{t_0}(x + iy) = 0$  with*

$$x \geq X + \sqrt{1 - y_0^2}, \quad y_0 \leq y \leq \sqrt{1 - 2t_0}.$$

(iii) (**Intermediate-time barrier.**) *There are no zeros  $H_t(x + iy) = 0$  with*

$$X \leq x \leq X + \sqrt{1 - y_0^2}, \quad \sqrt{y_0^2 + 2(t_0 - t)} \leq y \leq \sqrt{1 - 2t}, \quad 0 \leq t \leq t_0.$$

Then

$$\Lambda \leq t_0 + \frac{y_0^2}{2}.$$

Sections 3, 4–5, and 6 below describe the proof's suppliers for hypotheses (i), (ii), and (iii) respectively, at the parameters (4).

For hypothesis (i) one uses the normalization connecting  $H_0$  to  $\zeta$ : a zero  $H_0(x + iy) = 0$  gives  $\xi(\frac{1-y+ix}{2}) = 0$ ; the functional equation  $\xi(s) = \xi(1-s)$  followed by complex conjugation produces the representative zero  $\xi(\frac{1+y+ix}{2}) = 0$  with  $\sigma = \frac{1+y}{2} \geq \frac{1+y_0}{2}$  and ordinate  $T = \frac{x}{2}$ , which is the region excluded in (i). (The functional equation and the reality symmetry  $\overline{\xi(\bar{s})} = \xi(s)$  are the classical inputs; the affine algebra of this “sign map” is elementary.)

### 2.2 The effective approximation

For  $t > 0$  write, following [8],

$$g_t(z) = \frac{H_t(z)}{B_t(z)}, \tag{6}$$

where the normalizing factor is as follows. Let  $\text{Log}$  be the standard branch of the logarithm, with branch cut  $(-\infty, 0]$  and imaginary part in  $(-\pi, \pi]$ , and on  $\mathbb{C} \setminus (-\infty, 1]$  define

$$M_0(s) = \frac{\sqrt{2\pi}}{16} s(s-1)\pi^{-s/2} \exp\left(\left(\frac{s}{2} - \frac{1}{2}\right) \text{Log} \frac{s}{2} - \frac{s}{2}\right), \tag{7}$$

$$\alpha(s) = (\log M_0)'(s) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2} \text{Log} \frac{s}{2\pi}, \tag{8}$$

$$M_t(s) = \exp\left(\frac{t}{4}\alpha(s)^2\right) M_0(s), \quad B_t(x + iy) = M_t\left(\frac{1+y-ix}{2}\right). \tag{9}$$

These are equations (6), (8)–(11) of [8]. In particular  $M_0$ , and hence  $M_t$ , takes values in  $\mathbb{C} \setminus \{0\}$  on its domain; since  $(1+y-ix)/2$  is non-real whenever  $x \neq 0$ , it avoids the branch cut. Therefore  $B_t(x+iy) \neq 0$  for all  $t \geq 0$ ,  $x > 0$  and  $y > 0$ , as noted explicitly in [8, §1, after (11)]. This nonvanishing is used below both in the reduction of zero-freeness of  $H_t$  to lower bounds on  $|f_t|$  and in the argument-principle step of the barrier argument (Section 6). Theorem 1.3 of [8] provides, in the region

$$x \geq 200, \quad 0 \leq y \leq 1, \quad 0 < t \leq \frac{1}{2}, \quad (10)$$

an approximation

$$g_t(x+iy) = f_t(x+iy) + O_{\leq}(e_A + e_B + e_{C,0}), \quad (11)$$

where  $O_{\leq}(E)$  denotes a quantity of absolute value at most  $E$ , the main term is the finite sum (equation (14) of [8])

$$f_t(x+iy) = \sum_{n=1}^N b_t(n) n^{-s_*} + \gamma \sum_{n=1}^N n^y b_t(n) n^{-\overline{s_*} - \kappa}, \quad b_t(n) = \exp\left(\frac{t}{4} \log^2 n\right), \quad (12)$$

(we write  $b_t(n)$  for the quantity denoted  $b_n^t$  in [8]; all other symbols in (12) agree with the notation there, including the placement of  $\kappa$  alongside the conjugated exponent  $\overline{s_*}$  in the second sum) with:

- the window index  $N = \lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \rfloor$ ;
- the exact quantities (equations (16)–(18) of [8])

$$\gamma = \frac{M_t\left(\frac{1-y+ix}{2}\right)}{M_t\left(\frac{1+y-ix}{2}\right)}, \quad (13)$$

$$s_* = \frac{1+y-ix}{2} + \frac{t}{2} \alpha\left(\frac{1+y-ix}{2}\right), \quad (14)$$

$$\kappa = \frac{t}{2} \left[ \alpha\left(\frac{1-y+ix}{2}\right) - \alpha\left(\frac{1+y+ix}{2}\right) \right], \quad (15)$$

which satisfy, by equations (20)–(22) of [8], the bounds

$$|\gamma| \leq e^{0.02y} \left(\frac{x}{4\pi}\right)^{-y/2}, \quad (16)$$

$$\operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t}{4} \log \frac{x}{4\pi} - \frac{t}{2x^2} \left(1 - 3y + \frac{4y(1+y)}{x^2}\right)_+, \quad (17)$$

$$|\kappa| \leq \frac{ty}{2(x-6)}; \quad (18)$$

- and error terms  $e_A, e_B, e_{C,0}$  with explicit majorants (equations (71)–(72) and (23)–(24) of [8]).

Since  $B_t \neq 0$ , nonvanishing of  $H_t$  on a region follows from the strict inequality  $|f_t| > e_A + e_B + e_{C,0}$  there; this implication is itself stated in [8] as Corollary 1.4 (“Criterion for non-vanishing”).

### 2.3 The conservative error constant 10.50

Our proof deliberately does *not* use the constant 10.44 displayed in equation (24) of [8] for the majorization of  $e_{C,0}$ . Instead it derives a slightly weaker bound directly from Proposition 6.6(vi) of [8], which contains the expression

$$\exp\left(\frac{3|\log \frac{x}{4\pi} + \frac{i\pi}{2}| + 3.58}{x - 8.52}\right) \left(1 + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{6.92}{x - 12}\right).$$

On the theorem domain  $x \geq 200$  one has  $x - 8.52 > x - 12 > 0$ ; applying  $1 + u \leq e^u$  to the second factor and replacing the denominator  $x - 8.52$  by the smaller denominator  $x - 12$  yields the *conservative corollary*

$$e_{C,0} \leq \left(\frac{x}{4\pi}\right)^{-(1+y)/4} \exp\left(-\frac{t}{16} \log^2 \frac{x}{4\pi} + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{3|\log \frac{x}{4\pi} + \frac{i\pi}{2}| + 10.50}{x - 12}\right), \quad (19)$$

with  $3.58 + 6.92 = \frac{21}{2} = 10.50$  exactly. Every numerical error computation below (finite range, tail, and barrier) uses (19), and uses the denominator  $x - 6.66$ , which appears in equation (23) and in Proposition 6.6(iv),(v) of [8], for  $e_A + e_B$ .

**Remark 2.2.** Equation (24) of [8] displays the bound (19) with the constant 10.44 in place of 10.50. The sentence following Proposition 6.6 in [8] states that (24) is obtained from part (vi) of the proposition by “the inequality  $1 + u \leq \exp(u)$ ” followed by the bound  $\frac{1}{x-8.52} \leq \frac{1}{x-12}$ ; carrying out this recipe yields the constant  $3.58 + 6.92 = 10.50$ , and we are not aware of an argument in [8] producing the displayed 10.44. Accordingly, our proof does not rely on (24): all error computations use only (19), which follows rigorously from Proposition 6.6(vi) by the two elementary steps just described, uniformly on the domain (10). (The weaker constant costs a factor  $e^{0.06/(x-12)}$ , which is utterly negligible at  $x \approx 6 \times 10^{12}$ .)

## 3 Hypothesis (i): the verified height

Platt and Trudgian [7, Theorem 1] proved that every nontrivial zero of  $\zeta$  with ordinate in  $(0, T_{PT}]$ ,

$$T_{PT} = 3000175332800,$$

lies on the critical line; their computation accounts for the lowest 12 363 153 437 138 nontrivial zeros. Counting sign changes of the completed zeta function on the critical line and using Turing’s method to confirm completeness yields the off-line exclusion form used here. Hypothesis (i) of Theorem 2.1 requires the absence of zeros with  $\frac{1+y_0}{2} \leq \sigma \leq 1$  and  $0 \leq T \leq X/2$ . Since

$$\frac{X}{2} = 3000000092913.5, \quad T_{PT} - \frac{X}{2} = \frac{350479773}{2} > 0,$$

the verified region covers all ordinates  $0 < T \leq X/2$  with margin  $\approx 1.75 \times 10^8$ . On the segment  $T = 0$ ,  $\frac{1}{2} < \sigma < 1$ , one argues directly: the Dirichlet eta function satisfies  $\eta(\sigma) = \sum_{n \geq 1} (-1)^{n-1} n^{-\sigma} > 0$  for  $0 < \sigma < 1$  (group consecutive terms), while  $\eta(\sigma) = (1 - 2^{1-\sigma})\zeta(\sigma)$  with  $1 - 2^{1-\sigma} < 0$ ; hence  $\zeta(\sigma) < 0 \neq 0$  there. At  $\sigma = 1$ ,  $\zeta$  has a pole. This supplies hypothesis (i).

As with any use of a computationally established theorem, this step inherits the correctness of the Platt–Trudgian computation itself (a rigorous interval-arithmetic computation, independent of anything in the present paper); we use it as a published-theorem input.

## 4 Hypothesis (ii), finite range: the window scan

Hypothesis (ii) concerns nonvanishing of  $H_{t_0}$  on

$$x \geq x_* := X + \sqrt{1 - y_0^2}, \quad y_0 \leq y \leq \sqrt{1 - 2t_0}. \quad (20)$$

The proof splits this region at the abscissa corresponding to the window index  $N_* = 3840000$ : the finite range  $690988 \leq N \leq 3840000$  is handled in this section by 3,149,013 individual interval computations, and the infinite range  $N \geq 3840000$  in Section 5 by a single analytic contraction argument. The two ranges overlap on a full window, eliminating endpoint risk.

### 4.1 Windows and the freeze lemma

At  $t = t_0$  define

$$x_N = 4\pi \left( N^2 - \frac{t_0}{16} \right), \quad W_N = [x_N, x_{N+1}),$$

so that  $x \in W_N$  iff the window index of  $x$  equals  $N$ . Exact rational Machin-type bounds for  $\pi$  and exact integer square-root brackets give

$$x_{690988} < x_* < x_{690989}, \quad (21)$$

with margins exceeding  $5.37 \times 10^6$  and  $1.19 \times 10^7$  respectively; thus  $x_*$  lies strictly inside  $W_{690988}$ , and the half-open windows tile  $[x_*, x_{3840001})$  with no gaps or overlaps.

The computations of Section 4.3 evaluate, for each  $N$ , the majorants of (16)–(18) *frozen at the left edge*  $x = x_N$  of the window. The following elementary lemma justifies this.

**Lemma 4.1** (Window freeze). *For  $y_0 \leq y \leq \sqrt{1 - 2t_0}$  define*

$$\begin{aligned} G(x, y) &= e^{y/50} \left( \frac{x}{4\pi} \right)^{-y/2}, \\ K(x, y) &= \frac{t_0 y}{2(x - 6)}, \\ \Sigma(x, y) &= \frac{1 + y}{2} + \frac{t_0}{4} \log \frac{x}{4\pi} - \frac{t_0}{2x^2} \left( 1 - 3y + \frac{4y(1 + y)}{x^2} \right)_+. \end{aligned}$$

*Then for every  $N$  in the finite range, every  $x \in W_N$  and every  $y$  in the stated range,*

$$G(x, y) \leq G(x_N, y), \quad K(x, y) \leq K(x_N, y), \quad \Sigma(x, y) \geq \Sigma(x_N, y).$$

*Proof.*  $\partial_x G = -\frac{y}{2x} G < 0$  and  $\partial_x K = -\frac{t_0 y}{2(x-6)^2} < 0$  give the first two inequalities. For the third, write  $h(x, y) = 1 - 3y + A(y)/x^2$  with  $A(y) = 4y(1 + y) \geq 0$ ; then  $h_x = -2A/x^3 \leq 0$ , so  $h_+$  is nonincreasing in  $x$ , and since  $x^{-2}$  is positive and decreasing, the product  $x^{-2}h_+$  is nonincreasing. The logarithmic term is strictly increasing, so  $\Sigma$  is strictly increasing in  $x$ , including across the kink  $h = 0$  (both one-sided derivatives there are positive).  $\square$

Note that  $e^{y/50} = e^{0.02y}$ , matching (16);  $G, K, \Sigma$  frozen at  $x_N$  are precisely the quantities  $g_N$  (bound for  $|\gamma|$ ),  $k_N$  (bound for  $|\kappa|$ ) and  $\sigma_N$  (lower bound for  $\operatorname{Re} s_*$ ) used in the definitions of Section 4.3.



## 4.2 The mollified lower bound for $|f_{t_0}|$

Fix a window  $N$ , a height  $y > 0$ , and a finite set of auxiliary primes  $\mathcal{P}$  with  $D = \prod_{p \in \mathcal{P}} p$ . Let  $g_N, k_N, \sigma_N$  be as above, and introduce the real *Euler mollifier*

$$E(s_*) = \prod_{p \in \mathcal{P}} (1 - b_{t_0}(p) p^{-s_*}) = \sum_{d|D} \lambda_d d^{-s_*}, \quad \lambda_d = (-1)^{\omega(d)} \prod_{p|d} b_{t_0}(p) \in \mathbb{R}.$$

Define the convolved coefficients (for  $1 \leq n \leq DN$ )

$$B_{N,n} = \sum_{\substack{d|D, d|n \\ n/d \leq N}} \lambda_d b_{t_0}(n/d), \quad A_{N,n} = \sum_{\substack{d|D, d|n \\ n/d \leq N}} \lambda_d b_{t_0}(n/d) (n/d)^y,$$

so  $B_{N,1} = A_{N,1} = 1$ , and set

$$\begin{aligned} M_N &= \sum_{d|D} |\lambda_d| d^{-\sigma_N}, \quad C_N = \sum_{m=2}^N b_{t_0}(m) (m^{k_N} - 1) m^{y-\sigma_N}, \\ L_N &= \frac{1 - g_N - \sum_{n=2}^{DN} (|B_{N,n}| + g_N |A_{N,n}|) n^{-\sigma_N}}{M_N} - g_N C_N. \end{aligned} \quad (22)$$

**Lemma 4.2** (Mollified lower bound). *If  $L_N > 0$ , then  $|f_{t_0}(x + iy)| \geq L_N$  for all  $x \in W_N$ .*

*Proof.* Write (12) as  $f_t = A + \gamma C_\kappa$ , where

$$A = \sum_{m \leq N} b_t(m) m^{-s_*}, \quad C_\kappa = \sum_{m \leq N} m^y b_t(m) m^{-\bar{s}_* - \kappa}.$$

Exact Dirichlet convolution gives the two identities

$$E(s_*) A = \sum_{n=1}^{DN} B_{N,n} n^{-s_*}, \quad \overline{E(s_*)} C_0 = \sum_{n=1}^{DN} A_{N,n} n^{-\bar{s}_*}, \quad (23)$$

where the second identity uses that the  $\lambda_d$  are *real* (so that  $\overline{E(s_*)} = \sum_d \lambda_d d^{-\bar{s}_*}$ ). By the triangle inequality and  $\operatorname{Re} s_* \geq \sigma_N$ ,

$$|EA| \geq 1 - \sum_{n=2}^{DN} |B_{N,n}| n^{-\sigma_N}, \quad |\overline{E} C_0| \leq 1 + \sum_{n=2}^{DN} |A_{N,n}| n^{-\sigma_N}.$$

For complex  $\kappa$  with  $|\kappa| \leq k_N$ ,

$$|m^{-\kappa} - 1| = |e^{-\kappa \log m} - 1| \leq e^{|\kappa| \log m} - 1 \leq m^{k_N} - 1, \quad (24)$$

whence  $|\overline{E}(C_\kappa - C_0)| \leq |E| C_N$ . Combining, and using  $|\gamma| \leq g_N$ ,

$$|E| |f_t| \geq Q_N - g_N |E| C_N, \quad Q_N := 1 - g_N - \sum_{n=2}^{DN} (|B_{N,n}| + g_N |A_{N,n}|) n^{-\sigma_N}.$$

If  $L_N > 0$  then  $Q_N = M_N(L_N + g_N C_N) > 0$ ; the displayed inequality then forces  $E \neq 0$  (otherwise  $0 \geq Q_N > 0$ ). Dividing by  $|E|$  and using  $|E| \leq M_N$  together with  $Q_N > 0$ ,

$$|f_t| \geq \frac{Q_N}{|E|} - g_N C_N \geq \frac{Q_N}{M_N} - g_N C_N = L_N. \quad \square$$

The point of Lemma 4.2 is that any per-window lower bound for  $L_N$  is *already* in the same normalized units as the additive error in (11); no further Euler-factor conversion intervenes.

### 4.3 The lower bounds for $L_N$

For each integer  $N$  with  $690988 \leq N \leq 3840000$ , define a set of auxiliary primes

$$\mathcal{P}(N) = \begin{cases} \{2, 3, 5, 7, 11\}, & 690988 \leq N \leq 728999, \\ \{2, 3, 5, 7\}, & 729000 \leq N \leq 818999, \\ \{2, 3, 5\}, & 819000 \leq N \leq 1027999, \\ \{2, 3\}, & 1028000 \leq N \leq 3840000, \end{cases} \quad (25)$$

and let  $L_N$  denote the quantity (22) formed with the prime set  $\mathcal{P}(N)$ , at  $t = t_0$  and  $y = y_0$ , with the constants  $g_N = G(x_N, y_0)$ ,  $k_N = K(x_N, y_0)$ ,  $\sigma_N = \Sigma(x_N, y_0)$  frozen at the window's left edge as in Lemma 4.1. (The number of terms in  $L_N$  grows with  $|\mathcal{P}(N)|$ ; the choice (25) uses the larger, more effective mollifiers only on the ranges of  $N$  where they are needed.)

**Proposition 4.3** (Lower bounds for  $L_N$ ). *For every integer  $690988 \leq N \leq 3840000$ ,*

$$L_N \geq T_{\min} = 791366 \times 10^{-12} > 0.$$

*More precisely, the minimum of  $L_N$  over each of the four ranges in (25) is bounded below as listed in Table 1; the smallest certified lower endpoint is  $T_{\min}$  and occurs in the row  $N = 690988$ .*

| $\mathcal{P}(N)$     | range of $N$                  | lower bound for $\min L_N$ |
|----------------------|-------------------------------|----------------------------|
| $\{2, 3, 5, 7, 11\}$ | $690988 \leq N \leq 728999$   | 0.000000791366             |
| $\{2, 3, 5, 7\}$     | $729000 \leq N \leq 818999$   | 0.000315112459             |
| $\{2, 3, 5\}$        | $819000 \leq N \leq 1027999$  | 0.000305788807             |
| $\{2, 3\}$           | $1028000 \leq N \leq 3840000$ | 0.000309285478             |

Table 1: The auxiliary prime sets (25) and, for each of the four ranges of  $N$ , the lower bound for  $\min_N L_N$  established by the computation described in the proof of Proposition 4.3.

*Proof.* The primary computer-assisted evidence is the sealed chain consisting of [src/lemm\\_a\\_sweep\\_p235711.c](#), the complete invocation schedule [scripts/run\\_full\\_sweep.sh](#), the compressed row certificates under [certificates/](#), the fail-closed checker [verifiers/verify\\_finite\\_and\\_binding.py](#), and the preserved verdict [logs/finite\\_and\\_binding.log](#). It supplies one directed lower-bound row for every integer  $690988 \leq N \leq 3840000$  and checks parameter binding, complete coverage, absence of gaps and duplicates, and positivity. The 3,149,013 sealed rows give the four conservative floors in Table 1; the smallest is  $791366 \times 10^{-12}$  in the row  $N = 690988$ .

A derived second-toolchain reconstruction is provided by [independent/prop43/prop4\\_3\\_proof.c](#), with its complete schedule in [independent/prop43/INVOCATIONS.md](#). It reads no stored certificate data and computes outward-rounded enclosures in two modes: a direct term-by-term evaluation of (22), and a production sweep using activation-frozen weights, rigorous Taylor moments, and an explicit Lagrange remainder. At 256-bit precision with FLINT/Arb 3.6.0, the two modes agree at the extremal index  $N = 690988$  and reproduce the global floor. The derived run prints the slightly stronger intermediate floors 0.000315112481 and 0.000305788816 in the second and third ranges; those values are not used in Table 1,

whose weaker values come from the authoritative sealed corpus. The reconstruction outputs are unsealed and are retained under [dan-reworking/code/prop43/runs/](#); the sealed [in dependent/prop43/](#) directory contains only the source and invocation schedule. Thus the reconstruction corroborates, but does not replace, the primary chain.  $\square$

#### 4.4 Transfer to the full height range

Proposition 4.3 concerns the single height  $y = y_0$ , while hypothesis (ii) requires nonvanishing for all  $y \in [y_0, \sqrt{1-2t_0}]$ . For fixed  $N$  write  $L_N(y)$  for the quantity (22) with its explicit  $y$ -dependence restored (the constants  $g_N, k_N, \sigma_N$  frozen in  $x$  as before, but evaluated at height  $y$ ; likewise the coefficients  $A_{N,n}$  and the sums  $M_N, C_N$ ), so that  $L_N = L_N(y_0)$ . Throughout this subsection  $t = t_0$  and  $N$  are fixed, and we abbreviate

$$q_N = N^2 - \frac{t_0}{16}, \quad g = \frac{1}{50} - \frac{1}{2} \log q_N (< 0), \quad \rho_N = \frac{t_0}{2(x_N - 6)},$$

so that  $g_N(y) = e^{yg}$  exactly and  $k_N(y) = \rho_N y$ . The bridge between Proposition 4.3 and the full height range is the following statement.

**Proposition 4.4** (Height transfer). *For every integer  $690988 \leq N \leq 3840000$  and every  $y \in [y_0, \sqrt{1-2t_0}]$ ,*

$$L_N(y) \geq L_N(y_0).$$

The proof establishes three monotonicity statements: as  $y$  increases over  $[y_0, \sqrt{1-2t_0}]$ , (a) the majorant sum

$$\mathcal{F}_{N,\mathcal{P}}(y) = g_N(y) + \sum_{n=2}^{DN} (|B_{N,n}| + g_N(y)|A_{N,n}(y)|) n^{-\sigma_N(y)}$$

subtracted in (22) is nonincreasing; (b) the mollifier bound  $M_N(y)$  is positive and nonincreasing; and (c) the correction term  $g_N(y)C_N(y) \geq 0$  is nonincreasing. Statement (a) is the delicate one: the crude majorant obtained by discarding the negative term  $g_N g|A|$  below is not monotone, and the proof must retain the cancellation between the  $A$ -side terms. We build it up in three lemmas and one computation.

**Lemma 4.5** (Slope of the frozen exponent). *For  $y_0 \leq y_1 \leq y_2 \leq \sqrt{1-2t_0}$ ,*

$$\sigma_N(y_2) - \sigma_N(y_1) \geq \frac{1}{2} (y_2 - y_1).$$

*Proof.*  $\sigma_N(y) = \frac{1+y}{2} + \frac{t_0}{4} \log q_N - \frac{t_0}{2x_N^2} (1 - 3y + \frac{4y(1+y)}{x_N^2})_+$ . The inner expression has  $y$ -derivative  $-3 + \frac{4(1+2y)}{x_N^2} < 0$  (as  $4(1+2y) \leq 12 < 3x_N^2$ ), so its positive part is nonincreasing in  $y$  and the subtracted term is nondecreasing; the explicit term  $\frac{1+y}{2}$  contributes slope  $\frac{1}{2}$ .  $\square$

**Lemma 4.6** (Monotone normalizer and correction). *On  $[y_0, \sqrt{1-2t_0}]$ : (b)  $M_N(y)$  is positive and nonincreasing; (c)  $g_N(y)C_N(y)$  is nonnegative and nonincreasing.*

*Proof.* (b) Each term  $|\lambda_d|d^{-\sigma_N(y)}$  is positive; for  $d \geq 2$  it is nonincreasing by Lemma 4.5, and the  $d = 1$  term is constant.

(c) Nonnegativity is clear ( $m^{k_N(y)} \geq 1$ ). Fix  $y_0 \leq y_1 < y_2 \leq \sqrt{1-2t_0}$  and put  $\Delta = y_2 - y_1$ . First,  $g_N(y_2) = g_N(y_1)e^{g\Delta}$  exactly. Next, for each  $2 \leq m \leq N$ , the term  $b_t(m)(m^{\rho_N y} - 1)m^{y-\sigma_N(y)}$  of  $C_N$  grows by at most the factor  $e^{(\frac{1}{2}\log m + \rho_N \log m + \frac{1}{y_0})\Delta}$ : the exponent  $y - \sigma_N(y)$  grows by at most  $\frac{1}{2}\Delta$  (Lemma 4.5); and by the mean value theorem applied to  $y \mapsto \log(m^{\rho_N y} - 1)$ , whose derivative is  $\rho_N \log m \frac{e^z}{e^z - 1}$  with  $z = \rho_N y \log m > 0$ , the bound  $\frac{e^z}{e^z - 1} \leq 1 + \frac{1}{z}$  gives a growth rate at most  $\rho_N \log m + \frac{1}{y} \leq \rho_N \log m + \frac{1}{y_0}$ . Since a sum of positive terms grows no faster than its fastest-growing term, and  $m \leq N$ ,

$$g_N(y_2)C_N(y_2) \leq g_N(y_1)C_N(y_1) \exp\left[\left(g + \frac{1}{2}\log N + \rho_N \log N + \frac{1}{y_0}\right)\Delta\right].$$

It remains to check that the rate is negative for every  $N$  in the range. First,  $g + \frac{1}{2}\log N = \frac{1}{50} - \frac{1}{2}\log \frac{q_N}{N}$  and  $\frac{q_N}{N} = N - \frac{t_0}{16N} > N - 1 \geq 690987 > e^{13.44}$  (as  $e^{13.44} < 442414 \times 1.5528 < 687000$ ), so  $g + \frac{1}{2}\log N < 0.02 - 6.72$ . Second,  $\frac{1}{y_0} < \frac{10^7}{1872719} < 5.3399$ , since  $y_0^2 = \frac{350708}{10^7} > \left(\frac{1872719}{10^7}\right)^2 = 0.03507076 \dots$  gives  $y_0 > \frac{1872719}{10^7}$ . Third, uniformly over the range,  $\rho_N \log N \leq \frac{t_0 \log(3.84 \times 10^6)}{2(x_{690988} - 6)} < \frac{0.17 \times 16}{10^{13}} < 10^{-12}$ . Hence the rate is at most  $0.02 + 5.3399 + 10^{-12} - 6.72 < -1.36 < 0$ .  $\square$

**Lemma 4.7** (Dini bound in divisor-cell form). *Let  $2 \leq n \leq DN$ ,  $L = \log n$ , and let  $\mathcal{D}_{N,n} = \{d \mid D : d \mid n, n \leq dN\}$  be the active divisor set (which does not depend on  $y$ ), assumed nonempty. With  $\ell_d = \log d$  and*

$$\begin{aligned} r_d(L) &= \exp\left\{\frac{t_0}{4}\left(\sum_{p|d} \log^2 p + \ell_d^2 - 2L\ell_d\right)\right\}, \\ C_0 &= \sum_{d \in \mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L), \\ C(y) &= \sum_{d \in \mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L) d^{-y}, \\ C'(y) &= - \sum_{d \in \mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L) d^{-y} \ell_d, \end{aligned}$$

one has  $B_{N,n} = b_t(n) C_0$ ,  $A_{N,n}(y) = b_t(n) e^{yL} C(y)$  and  $\frac{d}{dy} A_{N,n}(y) = b_t(n) e^{yL} (LC(y) + C'(y))$ , and the upper right Dini derivative of the corresponding summand of  $\mathcal{F}_{N,\mathcal{P}}$  satisfies

$$D_y^+ \left[ (|B_{N,n}| + g_N(y)|A_{N,n}(y)|) n^{-\sigma_N(y)} \right] \leq b_t(n) n^{-\sigma_N(y)} \left( G h - \frac{L}{2} |C_0| \right),$$

where  $G = G(y) = e^{y(g+L)}$  and  $h = h(y) = |LC(y) + C'(y)| + (g - \frac{L}{2})|C(y)|$ .

*Proof.* The identities follow from  $\lambda_d b_t(n/d) = (-1)^{\omega(d)} b_t(n) r_d(L)$  — indeed  $\log b_t(n/d) = \frac{t_0}{4}(L - \ell_d)^2 = \log b_t(n) + \frac{t_0}{4}(\ell_d^2 - 2L\ell_d)$  and  $\lambda_d = (-1)^{\omega(d)} \exp(\frac{t_0}{4} \sum_{p|d} \log^2 p)$  — together with  $(n/d)^y = e^{yL} d^{-y}$ ; note also  $g_N(y)e^{yL} = e^{y(g+L)} = G$ . Since  $A = A_{N,n}(y)$  is real-analytic in  $y$ ,

$$D_y^+ \{g_N |A|\} = \begin{cases} g_N(g|A| + \operatorname{sgn}(A)A'), & A \neq 0, \\ g_N|A'|, & A = 0, \end{cases} \leq g_N(g|A| + |A'|)$$

in both cases. Both factors of the summand are nonnegative and locally Lipschitz,  $|B|$  is constant in  $y$ , and by Lemma 4.5,  $D_y^+ n^{-\sigma_N(y)} \leq -\frac{L}{2} n^{-\sigma_N(y)}$ ; the product rule for upper Dini

derivatives of nonnegative Lipschitz functions then gives

$$D_y^+ [(|B| + g_N |A|) n^{-\sigma_N}] \leq n^{-\sigma_N} \left( g_N (g|A| + |A'|) - \frac{L}{2} (|B| + g_N |A|) \right).$$

Substituting the identities and factoring out  $b_t(n) > 0$  yields the display.  $\square$

**Lemma 4.8** (Cell decomposition and  $q$ -freezing). (i) For  $2 \leq n \leq DN$  put  $G_0 = \gcd(n, D)$  and list the divisors of  $G_0$  as  $1 = d_0 < d_1 < \dots < d_s$  (so  $s+1 = \tau(G_0)$ ), with  $d_{-1} = 0$ . On the ratio sector  $d_{j-1} < n/N \leq d_j$  ( $0 \leq j \leq s$ ) the active set is exactly  $\mathcal{D}_{N,n} = \{d_j, \dots, d_s\}$ , and for  $n/N > d_s$  it is empty (the summand vanishes identically). The number of pairs (mask, sector) with nonempty active set is  $\sum_{G_0|D} \tau(G_0) = 3^{|\mathcal{P}|}$ .

(ii) With  $c = |C(y)|$  and  $K = |LC(y) + C'(y)| - \frac{L}{2}c$ , so that  $Gh_+ = U(g) := e^{y(L+g)} [K+gc]_+$ , the function  $\tilde{g} \mapsto U(\tilde{g})$  is nondecreasing.

(iii) On the sector of (i), within the range  $N_{\min} \leq N \leq N_{\max}$  of the prime set,  $q_N \geq \max(N_{\min}^2, e^{2L}/d_j^2) - \frac{t_0}{16}$ ; since  $g$  is decreasing in  $q_N$ , replacing  $g$  by its value at this smallest feasible  $q_N$  can only increase  $U(g)$ .

*Proof.* (i)  $d \in \mathcal{D}_{N,n}$  iff  $d | G_0$  and  $n \leq dN$ , i.e.  $d \geq n/N$ ; on the stated sector these are exactly  $d_j, \dots, d_s$ . Summing  $\tau(G_0)$  over the divisors  $G_0$  of the squarefree  $D$  gives  $\prod_{p \in \mathcal{P}} (\tau(1) + \tau(p)) = 3^{|\mathcal{P}|}$ . (ii) Where  $K + \tilde{g}c > 0$ ,  $U'(\tilde{g}) = e^{y(L+\tilde{g})} (y(K + \tilde{g}c) + c) \geq 0$ ; where  $K + \tilde{g}c \leq 0$ ,  $U = 0$ ; and  $U$  is continuous. (iii)  $n \leq d_j N$  gives  $N \geq e^L/d_j$ , and  $N \geq N_{\min}$ ;  $q_N$  is increasing in  $N$  and  $g$  decreasing in  $q_N$ .  $\square$

By Lemma 4.8, for each of the four prime sets the quantity  $Gh_+$  is bounded by an expression depending only on the cell, on  $L$  and on  $y$ ; the following finite family of inequalities is therefore checkable by a finite computation.

**Proposition 4.9** (Cell inequalities). For each of the four prime sets of (25), every  $N$  in the corresponding range, every  $2 \leq n \leq DN$  with  $\mathcal{D}_{N,n} \neq \emptyset$ , and every  $y \in [y_0, \sqrt{1-2t_0}]$ , in the notation of Lemma 4.7,

$$Gh_+ \leq \frac{L}{2} |C_0|.$$

*Proof.* The primary sealed evidence is [verifiers/verify\\_triangle\\_y\\_dini\\_arb.c](#) together with the preserved transcripts [logs/triangle\\_y\\_dini\\_180.log](#) and [logs/triangle\\_y\\_dini\\_256.log](#). The sealed verifier checks the full cell cover and the strict threshold used below. The derived second-toolchain program [dan-reworking/code/prop49/prop49\\_proof.c](#) is unsealed, reads no stored data, and fails closed at every decision; its role is corroborative.

For each prime set, the program enumerates the  $3^{|\mathcal{P}|}$  (mask, ratio-sector) cells of Lemma 4.8(i): for each divisor mask  $G_0 | D$  and each divisor  $d_j$  of  $G_0$ , the active set  $\{d_j, \dots, d_s\}$  with the sector constraint  $d_{j-1} < n/N \leq d_j$ . The  $(L, y)$ -domain of the cell, namely

$$L \in \begin{cases} [\log 2, \log(N_{\max} d_0)], & j = 0, \\ [\max(\log 2, \log(N_{\min} d_{j-1})), \log(N_{\max} d_j)], & j \geq 1, \end{cases}$$

and  $y \in [y_0, \sqrt{1-2t_0}]$ , is covered by closed binary64 rectangles whose endpoints are padded outward by  $2 \times 10^{-12}$ ; the containment of each exact logarithm in its padded double endpoint, and of the exact  $y_0$  and  $\sqrt{1-2t_0}$  in the fixed  $y$ -box, is itself checked by ball arithmetic before use. On each rectangle the program freezes  $q_N$  at the smallest feasible value

$\max(N_{\min}^2, e^{2L}/d_j^2) - \frac{t_0}{16}$  (splitting the  $L$ -range at the transition point with overlapping padding; conservative by Lemma 4.8(ii)–(iii)), forms ball enclosures of  $C_0$ ,  $C(y)$ ,  $C'(y)$  from the exact identity  $r_d(L) = e^{\frac{t_0}{4}(\sum_{p|d} \log^2 p + \ell_d^2)} e^{-\frac{t_0}{2}L\ell_d}$ , and establishes on each leaf either  $h \leq 0$  or the strict ball inequality  $G h_+ < \frac{L}{2}|C_0|$ , bisecting the longer side adaptively (depth at most 42) and failing closed on any unresolved leaf. Before starting, the program also checks numerically, at the smallest window, the two elementary inequalities  $4(1+2y) < 3x_{690988}^2$  and  $g < 0$ : the first is the assumption underlying Lemma 4.5, and the second makes the standalone term  $g_N(y) = e^{yg}$  decreasing in the proof of Proposition 4.4. Both have already been proved analytically in the proofs just cited, and the numerical check adds nothing to the present proof; it is a deliberate redundancy of the program, which refuses to run when these inequalities fail, so that a passing run remains meaningful even if the parameters hard-coded in the program were ever modified.

The program is run at both 180- and 256-bit precision; both runs establish all four prime sets with no unresolved leaves (297490, 52239, 9290 and 237 leaves respectively, identical covers at both precisions), and the aggregated ratio bound

$$\sup \frac{G h_+}{\frac{L}{2}|C_0|} \leq 0.99999860767275095 < 0.9999988 < 1,$$

the closing comparison being made against the fixed threshold 0.9999988. The worst leaf lies in the five-prime range at  $L \approx 1.042$ – $1.054$ ,  $y \approx 0.1997$ – $0.2022$ ; its margin  $1 - \text{ratio} \approx 1.4 \times 10^{-6}$  is the second-thinnest in the entire proof. Each run completes in under ten seconds; the reports are recorded as [dan-reworking/code/prop49/runs/prop49\\_report\\_180.txt](#) and [dan-reworking/code/prop49/runs/prop49\\_report\\_256.txt](#).  $\square$

*Proof of Proposition 4.4. Statement (a).* For each  $n$  with nonempty active set, Lemma 4.7 and Proposition 4.9 give  $D_y^+ \leq b_t(n)n^{-\sigma_N}(G h - \frac{L}{2}|C_0|) \leq 0$  on  $[y_0, \sqrt{1-2t_0}]$  (if  $h \leq 0$  both terms are nonpositive; otherwise apply the proposition). Summands with empty active set vanish identically, and the standalone term  $g_N(y) = e^{yg}$  is decreasing. Each summand is continuous, and a continuous function with nonpositive upper right Dini derivative on an interval is nonincreasing (if  $f(y_2) > f(y_1)$ , put  $\varepsilon = \frac{f(y_2)-f(y_1)}{2(y_2-y_1)}$  and let  $y^*$  be the supremum of the set where  $f(y) \leq f(y_1) + \varepsilon(y - y_1)$ ; then  $y^* < y_2$  and  $D^+f(y^*) \geq \varepsilon > 0$ ). Hence  $\mathcal{F}_{N,\mathcal{P}}$  is nonincreasing.

*Conclusion.* By Proposition 4.3,  $1 - \mathcal{F}_{N,\mathcal{P}}(y_0) = M_N(y_0)L_N(y_0) + M_N(y_0)g_N(y_0)C_N(y_0) \geq M_N(y_0)T_{\min} > 0$ . By (a) the numerator  $1 - \mathcal{F}_{N,\mathcal{P}}(y)$  is nondecreasing, hence positive on the whole interval; by Lemma 4.6(b) the denominator  $M_N(y)$  is positive and nonincreasing; so the quotient  $(1 - \mathcal{F}_{N,\mathcal{P}}(y))/M_N(y)$  is nondecreasing. Subtracting the nonnegative nonincreasing correction  $g_N(y)C_N(y)$  (Lemma 4.6(c)) preserves this, and  $L_N(y) \geq L_N(y_0)$  follows.  $\square$

## 4.5 Error bounds and the finite-range conclusion

The remaining ingredient is a uniform bound on the errors of Theorem 1.3 over the finite region.

**Proposition 4.10** (Uniform error bound, finite region (candidate)). *For all  $x \geq x_*$  and  $y_0 \leq y \leq \sqrt{1-2t_0}$ , the error terms of Theorem 1.3 at  $t = t_0$ , bounded using equation (23) of [8] for  $e_A + e_B$  and the conservative corollary (19) for  $e_{C,0}$ , satisfy*

$$e_A + e_B \leq 2.06 \times 10^{-12}, \quad e_{C,0} \leq 0.000000233492848188649183,$$

and

$$e_A + e_B + e_{C,0} \leq E_{\max} := 0.000000233494905212337849.$$

(The third bound is established directly and is sharper than the sum of the two displayed bounds, which are rounded upward for display.)

*Proof.* The primary sealed evidence is [verifiers/verify\\_finite\\_and\\_binding.py](#) together with [logs/finite\\_and\\_binding.log](#). The promoted program [independent/prop410/prop410\\_proof.py](#), sealed and executed by the quick verifier at snapshot [034a37acce703e128e63e2f36520b0c8af6c1c76](#), reads no stored certificate data. Both Python files import the same `mpmath.iv` backend, pinned to `mpmath==1.4.1` by the repository environment, set it to 220-bit precision, and use the same effective-error routine; the promoted file is therefore a same-backend refactoring, not an independent interval implementation. They evaluate majorants of equation (23) of [8] and of the corollary (19). Their only inputs are the exact rational parameters  $t_0$ ,  $y_0^2$  and  $N_0 = 690988$  (the enclosures are computed over the rational interval  $[t_0, t_0 + 10^{-9}]$ , which contains  $t_0$ ); they read no stored error data, and every acceptance condition below is a strict inequality between exact rational endpoints, checked in exact arithmetic. Because `mpmath` describes this interval module as experimental, those exact endpoint comparisons do not by themselves validate the enclosure semantics of every transcendental operation. Thus the backend assumption is load-bearing for this proposition and remains an explicit release blocker.

The analytic reduction is stated next at the exact time  $t_0$ . In the two implementations,  $x_{N_0}(t)$ , the logarithm, and the defect are instead evaluated over the enclosing  $t$ -interval just stated, and a directed upper endpoint is used for the defect. Conditional on valid enclosures, that is a conservative strengthening of the exact calculation below.

A single evaluation at the smaller window edge

$$x_0 := x_{N_0} = 4\pi \left( N_0^2 - \frac{t_0}{16} \right) < x_*$$

(the strict inequality is (21)) dominates the whole region  $x \geq x_*$ ,  $y_0 \leq y \leq \sqrt{1 - 2t_0}$ ,  $t = t_0$ . The relevant error majorants are decreasing in  $x$ , so evaluation at  $x_0$  is conservative. Write  $\ell_0 = \log \frac{x_0}{4\pi} = \log(N_0^2 - \frac{t_0}{16})$ ,

$$\delta_1 = \frac{t_0}{4} \log \left( 1 - \frac{t_0}{16N_0^2} \right)^{-1} + \frac{t_0}{2x_0^2}, \quad \sigma_1 = \frac{1+y_0}{2} + \frac{t_0}{2} \log N_0 - \delta_1 = \frac{1+y_0}{2} + \frac{t_0}{4} \ell_0 - \frac{t_0}{2x_0^2}.$$

More generally, for  $N \geq N_0$  put

$$\delta_N := \frac{t_0}{4} \log \left( 1 - \frac{t_0}{16N^2} \right)^{-1} + \frac{t_0}{2x_N^2}, \quad \tilde{\sigma}_N := \frac{1+y_0}{2} + \frac{t_0}{2} \log N - \delta_N.$$

Thus  $\tilde{\sigma}_{N_0} = \sigma_1$ . Both summands of  $\delta_N$  decrease with  $N$ , so  $\delta_N \leq \delta_1$  and  $\tilde{\sigma}_N$  grows with  $\log N$  at rate at least  $t_0/2$ . The exponent estimate used in part (i) below also gives  $\operatorname{Re} s_* \geq \tilde{\sigma}_N$  throughout window  $W_N$ .

(i) *The exponent.* In (17) the positive part is at most 1: the program checks  $8/x_0^2 < 3y_0$ , so  $\frac{4y(1+y)}{x^2} \leq \frac{8y}{x^2} < 3y$  for all  $y \geq y_0$ , and  $\operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t_0}{4} \log \frac{x}{4\pi} - \frac{t_0}{2x^2} \geq \sigma_1$  throughout the region.

(ii) *The bound (19) for  $e_{C,0}$ .* Each factor is monotone in the favorable direction: the prefactor  $(\frac{x}{4\pi})^{-(1+y)/4}$  and the term  $-\frac{t_0}{16} \log^2 \frac{x}{4\pi}$  decrease in  $x$  (checked:  $\ell_0 > 2 > 0$ ) and in  $y$ ;



the factor  $3^y + 3^{-y}$  increases in  $y$ ;  $N - 0.125$  increases with the window; and the final quotient decreases in  $x$  because  $3 \frac{d}{dx} |\log \frac{x}{4\pi} + \frac{i\pi}{2}| \leq \frac{3}{x}$  while the numerator exceeds  $10.50 > 3$  (checked). Hence (19) evaluated at  $x = x_0$  and  $N = N_0$ , with  $y = y_0$  in the decreasing factors and  $y = \sqrt{1 - 2t_0}$  in  $3^y + 3^{-y}$ , bounds  $e_{C,0}$  on the whole region; this evaluation is the program's quantity `error_c0`.

(iii) *The bound (23) for  $e_A + e_B$ .* Since  $4\pi n^2 \leq 4\pi N^2 \leq x(1 - \frac{t_0}{16N^2})^{-1}$  for every  $n \leq N$ , the bounds (16) and (18) give, uniformly on the region,

$$1 + |\gamma|N^{|\kappa|}n^y \leq 1 + e^{0.02} \left(1 - \frac{t_0}{16N_0^2}\right)^{-1/2} \exp\left(\frac{t_0 \log N_0}{2(x_0 - 6)}\right)$$

(the quantity  $\log N/(x_N - 6)$  decreases in  $N$ ), while  $\log^2 \frac{x}{4\pi n^2} \leq \log^2 \frac{x}{4\pi}$  for  $n \leq N$  and the monotonicity of  $\log^2(\frac{x}{4\pi})/(x - 6.66)$  in  $x$  (checked:  $\ell_0 > 2$ ) bound the last factor of (23) by  $\exp(\frac{(t_0^2/16)\ell_0^2 + 0.626}{x_0 - 6.66}) - 1$ . For the middle factor,  $\sum_{n=1}^N b_{t_0}(n) n^{-\operatorname{Re} s_*} \leq 1 + \sum_{n=2}^{m_0} b_{t_0}(n) n^{-\sigma_1} + T_N$  with  $m_0 = 2000$ , where  $T_N := \sum_{m_0 < n \leq N} b_{t_0}(n) n^{-\tilde{\sigma}_N}$ . The summand  $\exp(\frac{t_0}{4} \log^2 n - \tilde{\sigma}_N \log n)$  decreases in  $n$  on  $n \leq N$  (checked:  $\frac{1+y_0}{2} - \delta_1 > 0$ ), so  $T_N \leq \int_{\log m_0}^{\log N} e^{\psi(L)} dL$  with  $\psi(L) = (1 - \tilde{\sigma}_N)L + \frac{t_0}{4}L^2$ ; since  $\psi$  is convex,  $T_N \leq (\log N - \log m_0) \max(e^{\psi(\log m_0)}, e^{\psi(\log N)})$ . Finally, the two rate inequalities

$$\frac{t_0}{2} \log m_0 > \left(\log \frac{N_0}{m_0}\right)^{-1}, \quad \frac{t_0}{2} \log N_0 - \frac{1 - y_0}{2} - \delta_1 > \left(\log \frac{N_0}{m_0}\right)^{-1},$$

imply that each of the two endpoint products  $(\log N - \log m_0)e^{\psi(\log m_0)}$  and  $(\log N - \log m_0)e^{\psi(\log N)}$  is nonincreasing in  $N$  for  $N \geq N_0$  (using the growth of  $\tilde{\sigma}_N$  just noted and  $\delta_N \leq \delta_1$ ), so their values at  $N = N_0$  dominate every window; both rate inequalities are checked by the Python routine. The product of the three bounds is its quantity `error_ab`.

The primary verifier checks the twelve domain and sign conditions and the coarse load-bearing gate  $E_{\max} < 234 \times 10^{-9}$ . The promoted file repeats those conditions and additionally gates the three sharper displayed bounds by exact rational comparison of the computed upper endpoints. Its run completes in under a second; a human-readable report is retained, unsealed, at [dan-reworking/code/prop410/runs/prop410\\_report.txt](#). These two same-backend checks support the candidate assertion but do not resolve the `mpmath.iv` enclosure question; an operation-level validation or an Arb reimplementaion is required before treating the proposition as certified. Conditional on valid enclosures for the invoked operations, the analytic reduction and strict gates above establish all three displayed bounds.  $\square$

We can now state the conclusion of the finite range.

**Proposition 4.11** (Nonvanishing at time  $t_0$ , finite range).  *$H_{t_0}(x + iy) \neq 0$  for all  $x_* \leq x < x_{3840001}$  and  $y_0 \leq y \leq \sqrt{1 - 2t_0}$ .*

*Proof.* Let  $x$  be in the stated range and let  $N$  be its window index, so that  $690988 \leq N \leq 3840000$  by (21), and  $x \in W_N$ . For  $y$  in the stated range, Lemma 4.2 (applied with the height  $y$ , the prime set  $\mathcal{P}(N)$ , and the frozen constants of Lemma 4.1), together with Propositions 4.3 and 4.4, gives

$$|f_{t_0}(x + iy)| \geq L_N(y) \geq L_N(y_0) \geq T_{\min}.$$

Combining with Proposition 4.10,

$$|f_{t_0}(x + iy)| - (e_A + e_B + e_{C,0}) \geq T_{\min} - E_{\max} \geq 0.000000557871094787 > 0. \quad (26)$$



The point  $(x, y, t_0)$  lies in the region (10), so [8, Corollary 1.4] (the criterion for non-vanishing quoted in Section 2.2) yields  $H_{t_0}(x + iy) \neq 0$ .  $\square$

## 5 Hypothesis (ii), infinite tail

This section proves nonvanishing of  $H_t$  for *all* windows  $N \geq N_* = 3840000$ , all  $t \in I_t = [\frac{129}{800}, \frac{161250001}{10^9}]$  (a closed interval whose left endpoint is exactly  $t_0$ ), and all  $y \in [y_0, \sqrt{1 - 2t}]$ , by a single interval computation at the cutoff  $N_*$  combined with monotonicity arguments in  $N$  and  $y$ . Since  $t_0 \in I_t$ , this in particular covers the case  $t = t_0$  needed for hypothesis (ii); the last window  $W_{3840000}$  of the finite range overlaps the tail's starting abscissa  $x_{3840000}$ , so the two ranges jointly cover all  $x \geq x_*$ .

### 5.1 Setup: mollifier and frozen exponents

Fix the sparse mollifier support  $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 10, 11, 13, 14\}$  and the exact rational vector

|             |   |                      |                      |                  |                      |                     |                      |                   |                    |                    |                         |
|-------------|---|----------------------|----------------------|------------------|----------------------|---------------------|----------------------|-------------------|--------------------|--------------------|-------------------------|
| $d$         | 1 | 2                    | 3                    | 4                | 5                    | 6                   | 7                    | 10                | 11                 | 13                 | 14                      |
| $\lambda_d$ | 1 | $-\frac{1021}{1000}$ | $-\frac{1054}{1000}$ | $-\frac{9}{200}$ | $-\frac{1119}{1000}$ | $\frac{1001}{1000}$ | $-\frac{1043}{1000}$ | $\frac{128}{125}$ | $-\frac{161}{200}$ | $-\frac{447}{500}$ | $\frac{456373}{500000}$ |

( $\lambda_1 = 1$ ;  $\sum_d |\lambda_d| = 9.918746$  exactly), and set  $M_\lambda(z) = \sum_{d \in \mathcal{S}} \lambda_d d^{-z}$ . (Unlike the Euler-product mollifier of Section 4.2, this vector was chosen by numerical optimization; its only structural requirement is  $\lambda_1 = 1$  and real entries.) Fix also the head lengths  $M = 153814$  and  $M_{\text{err}} = 3000$ .

For  $x \geq X_N(t) := 4\pi(N^2 - \frac{t}{16})$  with window index  $N \geq N_*$ , set

$$\delta(N, t) = \frac{t}{4} \left[ -\log\left(1 - \frac{t}{16N^2}\right) \right] + \frac{t}{2X_N(t)^2}, \quad k(N, t) = \frac{t}{2(X_N(t) - 6)}.$$

Both functions decrease in  $N$  for fixed  $t$ . Define the frozen constants exactly by the compact-interval maxima

$$\widehat{\delta} := \max_{t \in I_t} \delta(N_*, t), \quad \widehat{k} := \max_{t \in I_t} k(N_*, t).$$

They also increase in  $t$  on  $I_t$ , so these maxima equal their values at  $t = 161250001/10^9$ . The frozen exponents are then

$$\sigma_1 = \frac{1+y}{2} + \frac{t}{2} \log N - \widehat{\delta}, \quad \sigma_2 = \frac{1-y}{2} + \frac{t}{2} \log N - \widehat{\delta} - \widehat{k}, \quad G = e^{0.02y} \left( N^2 - \frac{t}{16} \right)^{-y/2}.$$

**Lemma 5.1** (Frozen exponents). *Let  $N \geq N_*$ ,  $t \in I_t$ ,  $0 < y \leq 1$ , and let  $x \in [X_N(t), X_{N+1}(t))$ , so that  $x$  has window index  $N$ . Then*

$$|\gamma| \leq G, \quad \operatorname{Re} s_* \geq \sigma_1, \quad |\kappa| \leq \widehat{k},$$

and consequently each term of the second sum in (12) satisfies

$$|n^y b_t(n) n^{-\overline{s_*} - \kappa}| \leq b_t(n) n^{-\sigma_2} \quad (1 \leq n \leq N).$$

*Proof.* Since  $x \geq X_N(t)$  we have  $x/4\pi \geq N^2 - t/16$ , and the three factors  $(x/4\pi)^{-y/2}$ ,  $x^{-2}$ ,  $(x-6)^{-1}$  appearing in (16)–(18) are decreasing in  $x$ , hence maximal at the window's left edge.

For  $\gamma$  this gives  $|\gamma| \leq e^{0.02y}(N^2 - t/16)^{-y/2} = G$  directly. For  $\kappa$ , using  $y \leq 1$  and that  $X_N(t)$  increases with  $N$ ,

$$|\kappa| \leq \frac{ty}{2(x-6)} \leq \frac{t}{2(X_N(t)-6)} = k(N, t) \leq k(N_*, t) \leq \widehat{k}.$$

For  $\operatorname{Re} s_*$ : by the exact identity

$$\frac{t}{4} \log\left(N^2 - \frac{t}{16}\right) = \frac{t}{2} \log N - \frac{t}{4} \left[ -\log\left(1 - \frac{t}{16N^2}\right) \right],$$

the logarithmic term of (17) is at least  $\frac{t}{2} \log N$  minus the first summand of  $\delta(N, t)$ . The subtracted positive-part term of (17) is at most  $t/(2x^2) \leq t/(2X_N(t)^2)$ , the second summand of  $\delta(N, t)$ : indeed, on the domain  $0 \leq y \leq 1$ ,  $x \geq 200$  one has  $1 - 3y + 4y(1+y)/x^2 \leq 1$  (for  $y > 0$  this is  $4(1+y) \leq 3x^2$ , and at  $y = 0$  the left side equals 1), so the positive part is at most 1. Both summands of  $\delta(N, t)$  decrease in  $N$  (in the first,  $t/16N^2$  decreases and  $u \mapsto -\log(1-u)$  is increasing; in the second,  $X_N(t)$  increases), so

$$\operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t}{2} \log N - \delta(N, t) \geq \frac{1+y}{2} + \frac{t}{2} \log N - \widehat{\delta} = \sigma_1.$$

Finally, for  $1 \leq n \leq N$ ,

$$|n^y b_t(n) n^{-\overline{s_*} - \kappa}| = b_t(n) n^{y - \operatorname{Re} s_* - \operatorname{Re} \kappa} \leq b_t(n) n^{y - \sigma_1 + \widehat{k}} = b_t(n) n^{-\sigma_2},$$

since  $-\operatorname{Re} \kappa \leq |\kappa| \leq \widehat{k}$  and  $\sigma_2 = \sigma_1 - y - \widehat{k}$ .  $\square$

## 5.2 The endpoint-cap lemma

The analytic engine for tail sums is:

**Lemma 5.2** (Endpoint cap). *For  $1 \leq a < c$  set  $E_{t,\sigma}(u) = \exp((1-\sigma) \log u + \frac{t}{4} \log^2 u)$  and*

$$\operatorname{Cap}_t(a, c; \sigma) = \max(E_{t,\sigma}(a), E_{t,\sigma}(c)) \log \frac{c}{a}.$$

*If  $\sigma > \frac{t}{2} \log c$ , then for all integers  $a \leq J < K \leq c$ ,*

$$\sum_{J < n \leq K} b_t(n) n^{-\sigma} \leq \operatorname{Cap}_t(a, c; \sigma).$$

*Proof.* The logarithmic derivative of  $b_t(u)u^{-\sigma}$  in  $\log u$  is  $\frac{t}{2} \log u - \sigma < 0$  on  $[a, c]$ , so the summand decreases and the sum is at most  $\int_a^c b_t(u)u^{-\sigma} du$ . Substituting  $v = \log u$  makes the integrand  $\exp((1-\sigma)v + \frac{t}{4}v^2)$ , which is log-convex in  $v$ , hence maximized at an endpoint of  $[\log a, \log c]$ ; multiplying by the length  $\log(c/a)$  of the  $v$ -interval gives the bound.  $\square$

## 5.3 Exact convolution and the contraction $D$

Define, for  $N \geq N_*$  and  $(x, y, t)$  with  $x$  in the window of index  $N$ ,

$$P = \sum_{m=2}^M |c_m(t)| m^{-\sigma_1}, \quad c_m(t) = \sum_{\substack{d \in \mathcal{S} \\ d|m}} \lambda_d b_t(m/d), \quad M_{\max} = M_{\max}(N, t, y) := \sum_{d \in \mathcal{S}} |\lambda_d| d^{-\sigma_1},$$

$$\text{TR} = \sum_{d \in \mathcal{S}} |\lambda_d| d^{-\sigma_1} \text{Cap}_t(\lfloor M/d \rfloor, N; \sigma_1), \quad \text{AB} = G\left(\sum_{k \leq M} b_t(k) k^{-\sigma_2} + \text{Cap}_t(M, N; \sigma_2)\right),$$

together with the nonnegative “overshoot” term

$$\text{OV} = \sum_{\substack{d \in \mathcal{S} \\ d > 1}} |\lambda_d| d^{-\sigma_1} \text{Cap}_t\left(\frac{N}{d+1}, N; \sigma_1\right)$$

(included for compatibility with the program of Proposition 5.10; see the remark after the proof), and set

$$D = D(N, t, y) := P + \text{TR} + \text{OV} + M_{\max} \text{AB}. \quad (27)$$

**Lemma 5.3** (Mollified contraction bound). *Assume the cap-validity conditions*

$$\hat{\delta} < \frac{1+y}{2}, \quad \hat{\delta} + \hat{k} < \frac{1-y}{2} \quad (28)$$

(equivalently  $\sigma_1 > \frac{t}{2} \log N$  and  $\sigma_2 > \frac{t}{2} \log N$ : the differences  $\sigma_i - \frac{t}{2} \log N$  are independent of  $N$ ). Then for every  $N \geq N_*$ ,  $t \in I_t$ ,  $0 < y \leq 1$  and  $x$  in the window of index  $N$ ,

$$|M_\lambda(s_*) f_t(x + iy) - 1| \leq D.$$

Consequently, if  $D < 1$  then

$$|f_t(x + iy)| \geq \frac{1-D}{M_{\max}}. \quad (29)$$

*Proof.* Write the two sums in (12) as  $F_B$  and  $F_A$  ( $f_t = F_B + \gamma F_A$ ). Expanding  $M_\lambda(s_*) F_B = \sum_{d \in \mathcal{S}} \sum_{k \leq N} \lambda_d b_t(k) (dk)^{-s_*}$  and partitioning the index pairs  $(d, k)$  by  $dk \leq M$  versus  $dk > M$  — a disjoint and exhaustive partition — gives the *exact* identity

$$M_\lambda(s_*) F_B - 1 = \sum_{m=2}^M c_m(t) m^{-s_*} + \sum_{d \in \mathcal{S}} \lambda_d d^{-s_*} \sum_{\lfloor M/d \rfloor < k \leq N} b_t(k) k^{-s_*}. \quad (30)$$

Here the first term groups the pairs with  $dk \leq M$  by  $m = dk$ : since  $m \leq M < N_* \leq N$ , the constraint  $k = m/d \leq N$  is automatic, so the coefficient of  $m^{-s_*}$  is exactly  $c_m(t)$ , and the  $m = 1$  term  $c_1 = \lambda_1 b_t(1) = 1$  has been moved to the left side. In the second term, for fixed  $d$  the condition  $dk > M$  for integer  $k$  is  $k > \lfloor M/d \rfloor$ . By  $\text{Re } s_* \geq \sigma_1$  (Lemma 5.1), the first term is bounded by  $P$ ; and each inner sum of the second term is bounded, by Lemma 5.2 with  $(a, c) = (\lfloor M/d \rfloor, N)$ , by  $\text{Cap}_t(\lfloor M/d \rfloor, N; \sigma_1)$ , so the second term is bounded by  $\text{TR}$ ; the application of Lemma 5.2 is legitimate because its hypothesis  $\sigma_1 > \frac{t}{2} \log N$  is the first condition of (28). On the  $A$ -side, Lemma 5.1 bounds each term of  $|\gamma F_A|$  by  $G b_t(k) k^{-\sigma_2}$ ; splitting at  $k = M$  and applying Lemma 5.2 with  $(a, c) = (M, N)$  at exponent  $\sigma_2$  (legitimate by the second condition of (28)) gives  $|\gamma F_A| \leq \text{AB}$ , while  $|M_\lambda(s_*)| \leq \sum_d |\lambda_d| d^{-\sigma_1} = M_{\max}$ ; hence  $|M_\lambda(s_*) \gamma F_A| \leq M_{\max} \text{AB}$ . Adding the nonnegative  $\text{OV}$  only weakens the bound, and the first claim follows. For the second claim: if  $D < 1$  then  $|M_\lambda(s_*) f_t| \geq 1 - D > 0$ ; in particular  $M_\lambda(s_*) \neq 0$ , and dividing by  $|M_\lambda(s_*)| \leq M_{\max}$  gives  $|f_t| \geq (1 - D)/M_{\max}$ .  $\square$

**Remark 5.4.** The term  $\text{OV}$  plays no role in the mathematics — the identity (30) is fully covered by  $P + \text{TR}$  — but the program of Proposition 5.10 below includes it in  $D$ , so it is included in (27) in order that the numerical bound on  $D$  majorizes the same quantity the program computes.

## 5.4 Reduction to the cutoff

The interval computation of Section 5.6 evaluates  $D$ , and the error majorants, only at the cutoff  $N = N_*$ , over the closed intervals

$$I_{\text{box}} = \left[ \frac{1872719}{10^7}, \frac{23409}{125000} \right], \quad I_{\text{ext}} = \left[ \frac{1872719}{10^7}, \frac{8231039}{10^7} \right],$$

where  $I_{\text{box}}$  contains  $y_0$  and the top of  $I_{\text{ext}}$  is at least  $\sqrt{1-2t}$  for every  $t \in I_t$ . The reduction from this single evaluation to all  $N \geq N_*$  and the full height range is carried out in Lemmas 5.5–5.9 below. Throughout, the *cutoff conditions* are the following inequalities, all evaluated at  $N = N_*$ , uniformly for  $t \in I_t$  and  $y \in I_{\text{ext}}$ :

$$\frac{t}{2} \log a (\log N_* - \log a) > 1 \quad \text{and} \quad (\sigma - 1)(\log N_* - \log a) > 1 \quad (31)$$

for each cap  $\text{Cap}_t(a, \cdot; \sigma)$  appearing in  $D$  and in the error bounds (the left endpoints  $a = \lfloor M/d \rfloor$  for  $d \in \mathcal{S}$ ,  $a = M$ , and  $a = M_{\text{err}}$ , with  $\sigma = \sigma_1$  or  $\sigma_2$  as appropriate), together with the  $A$ -side slope condition

$$0.02 - \frac{1}{2} \log(N_*^2 - \frac{t}{16}) + \frac{1}{2} \log N_* < 0. \quad (32)$$

**Lemma 5.5** (Decay of fixed-endpoint caps). *Let  $t > 0$ , let  $1 < a < N_*$ , let  $\sigma(q) = \sigma_0 + \frac{t}{2}q$  with  $\sigma_0 \in \mathbb{R}$ , and write  $q_* = \log N_*$ . Assume the two conditions (31) for this pair, that is,  $\frac{t}{2} \log a (q_* - \log a) > 1$  and  $(\sigma(q_*) - 1)(q_* - \log a) > 1$ . Then both conditions persist for all  $q \geq q_*$ , one has  $\sigma(q) > 1$  for all  $q \geq q_*$ , and*

$$q \mapsto \text{Cap}_t(a, e^q; \sigma(q))$$

*is positive and decreasing on  $[q_*, \infty)$ .*

*Proof.* By definition  $\text{Cap}_t(a, e^q; \sigma(q)) = \max(F_1(q), F_2(q))$ , where

$$F_1(q) = E_{t, \sigma(q)}(a) (q - \log a), \quad F_2(q) = E_{t, \sigma(q)}(e^q) (q - \log a).$$

Since  $\log E_{t, \sigma}(u) = (1 - \sigma) \log u + \frac{t}{4} \log^2 u$  and  $\sigma'(q) = \frac{t}{2}$ ,

$$\frac{d}{dq} \log F_1(q) = -\frac{t}{2} \log a + \frac{1}{q - \log a},$$

which is negative exactly when  $\frac{t}{2} \log a (q - \log a) > 1$ ; both factors on the left of this condition are positive and nondecreasing in  $q$ , so the condition, true at  $q_*$ , persists for all  $q \geq q_*$ . Similarly,

$$\frac{d}{dq} \log F_2(q) = \left[ (1 - \sigma(q)) - \frac{t}{2} q \right] + \frac{t}{2} q + \frac{1}{q - \log a} = 1 - \sigma(q) + \frac{1}{q - \log a},$$

which is negative exactly when  $(\sigma(q) - 1)(q - \log a) > 1$ . At  $q_*$  this condition forces  $\sigma(q_*) > 1$ ; then both factors  $\sigma(q) - 1$  and  $q - \log a$  are positive and increasing in  $q$ , so the condition persists, and  $\sigma(q) > 1$  throughout. Hence  $F_1$  and  $F_2$  are positive and decreasing on  $[q_*, \infty)$ , and so is their maximum.  $\square$

**Lemma 5.6** (Componentwise decay in  $N$ ). *Fix  $t \in I_t$  and  $y \in (0, 1]$ , and assume the cutoff conditions (31). Then each of the following is nonincreasing in  $N$  on  $[N_*, \infty)$  (throughout,  $q = \log N$  and the frozen exponents  $\sigma_i = \sigma_i(q)$  are increasing in  $q$  with slope  $\frac{t}{2}$ ):*

(i)  $m^{-\sigma_i(q)}$  for every fixed integer  $m \geq 2$  and  $i \in \{1, 2\}$ ; hence the head  $P$ , the mollifier bound  $M_{\max}$ , and the  $A$ -side head  $\sum_{k \leq M} b_t(k) k^{-\sigma_2}$  (whose  $k = 1$  term is constant) are nonincreasing;

(ii)  $G = e^{0.02y} (e^{2q} - \frac{t}{16})^{-y/2}$ ;

(iii) the moving-endpoint term  $OV$ .

*Proof.* (i) is immediate:  $\sigma_i(q)$  is increasing and  $\log m > 0$ , while the coefficients  $|c_m(t)|$ ,  $|\lambda_d|$  and  $b_t(k)$  do not depend on  $N$ . For (ii),

$$\frac{d}{dq} \log G = -\frac{y}{2} \cdot \frac{2e^{2q}}{e^{2q} - t/16} = -y \frac{N^2}{N^2 - t/16} < 0.$$

For (iii), each summand of  $OV$  is  $|\lambda_d| d^{-\sigma_1(q)} \text{Cap}_t(e^q/(d+1), e^q; \sigma_1(q))$ , a product of the nonincreasing factor  $d^{-\sigma_1(q)}$  ( $d \geq 2$ ) with a cap whose endpoints both scale with  $N$  and whose length  $\log(d+1)$  is constant. For  $r \in \{1, d+1\}$ ,

$$\frac{d}{dq} \left[ (1 - \sigma(q))(q - \log r) + \frac{t}{4}(q - \log r)^2 \right] = (1 - \sigma(q)) - \frac{t}{2}(q - \log r) + \frac{t}{2}(q - \log r) = 1 - \sigma(q) < 0,$$

using  $\sigma_1(q) > 1$  from Lemma 5.5. So both cap candidates are positive and decreasing, hence so are their maximum and the summand.  $\square$

**Lemma 5.7** (Decay in  $y$  at the cutoff). *Fix  $t \in I_t$  and assume the  $A$ -side slope condition (32). Then  $y \mapsto D(N_*, t, y)$  and  $y \mapsto M_{\max}(N_*, t, y)$  are nonincreasing on  $(0, 1]$ .*

*Proof.* Recall  $\partial \sigma_1 / \partial y = \frac{1}{2}$  and  $\partial \sigma_2 / \partial y = -\frac{1}{2}$ . On the  $B$ -side, every  $y$ -dependent factor has the form  $u^{-\sigma_1(y)}$  or  $E_{t, \sigma_1(y)}(u)$  with  $u > 1$  (the head terms  $m \geq 2$ , the mollifier terms  $d \geq 2$ , and the cap ingredients at  $u = \lfloor M/d \rfloor$ ,  $u = N_*$  or  $u = N_*/r$ , all exceeding 1), and satisfies  $\partial_y \log(\cdot) = -\frac{1}{2} \log u < 0$ ; the remaining factors ( $|c_m(t)|$ ,  $|\lambda_d|$ ,  $b_t$ , cap lengths) are independent of  $y$ , and maxima of nonincreasing functions are nonincreasing. Hence  $P$ ,  $TR$ ,  $OV$  and  $M_{\max}$  are nonincreasing in  $y$ . On the  $A$ -side,  $\log G$  is linear in  $y$  with slope  $0.02 - \frac{1}{2} \log(N_*^2 - \frac{t}{16})$ , so each head term and each cap ingredient of  $AB$ , with  $u \leq N_*$ , satisfies

$$\partial_y \log(G b_t(u) u^{-\sigma_2(y)}) \text{ resp. } \partial_y \log(G E_{t, \sigma_2(y)}(u)) = 0.02 - \frac{1}{2} \log(N_*^2 - \frac{t}{16}) + \frac{1}{2} \log u,$$

which is at most the left side of (32), hence negative. So  $AB$  is nonincreasing in  $y$ , and therefore so are  $M_{\max}AB$  and  $D$ .  $\square$

**Lemma 5.8** (Reduction to the cutoff). *Assume the cutoff conditions (31)–(32) hold. Then for every  $N \geq N_*$ , every  $t \in I_t$  and every  $y \in [y_0, \sqrt{1 - 2t}]$ ,*

$$D(N, t, y) \leq \sup_{y' \in I_{\text{box}}} D(N_*, t, y'), \quad M_{\max}(N, t, y) \leq \sup_{y' \in I_{\text{box}}} M_{\max}(N_*, t, y').$$

*Proof.* Fix  $t$  and  $y$ . Every constituent of  $D = P + TR + OV + M_{\max}AB$  is a sum of products of nonnegative factors, each nonincreasing in  $N$  on  $[N_*, \infty)$ : the heads,  $M_{\max}$  and  $G$  by Lemma 5.6(i)–(ii); the fixed-endpoint caps in  $TR$  and  $AB$  by Lemma 5.5, applied with  $a = \lfloor M/d \rfloor$  (at  $\sigma_1$ ) and  $a = M$  (at  $\sigma_2$ ), whose conditions (31) hold by hypothesis; and  $OV$  by

Lemma 5.6(iii). Sums and products of nonnegative nonincreasing functions are nonincreasing, so  $D(N, t, y) \leq D(N_*, t, y)$  and  $M_{\max}(N, t, y) \leq M_{\max}(N_*, t, y)$ . If  $y \in I_{\text{box}}$  the claims follow at once. Otherwise  $y > \max I_{\text{box}}$ , since  $y \geq y_0 \geq \min I_{\text{box}}$ ; as  $y \leq \sqrt{1-2t} < 1$ , Lemma 5.7 applies on  $[\max I_{\text{box}}, y]$  and gives  $D(N_*, t, y) \leq D(N_*, t, \max I_{\text{box}}) \leq \sup_{y' \in I_{\text{box}}} D(N_*, t, y')$ , and the same for  $M_{\max}$ .  $\square$

## 5.5 Tail error terms

The error bounds use the exact definitions (71)–(72) of [8] rather than the coarser display (23): (23) applies the worst-case factor  $N^{|\kappa|}$  to every term, whereas (71) carries the per-term factor  $n^{-\Re \kappa}$ , which Lemma 5.1 bounds by  $n^{\hat{k}}$ . Define, for  $N \geq N_*$ ,  $t \in I_t$  and  $0 < y \leq 1$ ,

$$L_N(t) = \log\left(N^2 - \frac{t}{16}\right), \quad \Delta(N, t) = \frac{\frac{t^2}{16} L_N(t)^2 + 0.626}{X_N(t) - 6.66},$$

and the two majorants

$$\begin{aligned} E_{AB}(N, t, y) &= (e^{\Delta(N, t)} - 1)(S_1(N, t, y) + G S_2(N, t, y)), \\ S_i(N, t, y) &= \sum_{n=1}^{M_{\text{err}}} b_t(n) n^{-\sigma_i} + \text{Cap}_t(M_{\text{err}}, N; \sigma_i) \quad (i = 1, 2), \\ E_C(N, t, y) &= \left(N^2 - \frac{t}{16}\right)^{-(1+y)/4} \\ &\quad \times \exp\left(-\frac{t}{16} L_N(t)^2 + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{3\sqrt{L_N(t)^2 + \pi^2/4} + 10.50}{X_N(t) - 12}\right). \end{aligned}$$

**Lemma 5.9** (Error majorants). *Assume the cap-validity conditions (28) and the conditions (31) for the caps with left endpoint  $a = M_{\text{err}}$ . Then for every  $N \geq N_*$ ,  $t \in I_t$ ,  $y \in [y_0, \sqrt{1-2t}]$  and  $x$  in the window of index  $N$ ,*

$$e_A + e_B \leq E_{AB}(N, t, y) \leq E_{AB}(N_*, t, y), \quad e_{C,0} \leq E_C(N, t, y) \leq E_C(N_*, t, y).$$

*Proof.* The per-term factor. The quantity  $\varepsilon_{t,n}$  of [8, (44)] is bounded, by the display in the proof of [8, Proposition 6.6(iv),(v)],

$$\varepsilon_{t,n}\left(\frac{1 \pm y + ix}{2}\right) \leq \exp\left(\frac{\frac{t^2}{32} \log^2 \frac{x}{4\pi n^2} + 0.313}{\frac{x}{2} - 3.33}\right) - 1 = \exp\left(\frac{\frac{t^2}{16} \log^2 \frac{x}{4\pi n^2} + 0.626}{x - 6.66}\right) - 1,$$

the imaginary part of  $\frac{1 \pm y + ix}{2}$  being  $\frac{x}{2}$ . For  $1 \leq n \leq N$  and  $x$  in the window of index  $N$  one has  $\frac{x}{4\pi n^2} \leq \frac{x}{4\pi}$  and  $\frac{x}{4\pi n^2} \geq \frac{X_N(t)}{4\pi N^2} = 1 - \frac{t}{16N^2}$ , so

$$\left|\log \frac{x}{4\pi n^2}\right| \leq \max\left(\log \frac{x}{4\pi}, -\log\left(1 - \frac{t}{16N^2}\right)\right) = \log \frac{x}{4\pi},$$

the last step because  $\frac{x}{4\pi}\left(1 - \frac{t}{16N^2}\right) \geq \frac{(N^2 - t/16)^2}{N^2} \geq 1$  (indeed  $N^2 - \frac{t}{16} \geq N$  as  $N \geq N_*$ ). Hence the per-term exponent is at most  $\varphi(x) = (\frac{t^2}{16} \log^2 \frac{x}{4\pi} + 0.626)/(x - 6.66)$ . Writing  $L = \log \frac{x}{4\pi}$ ,

which exceeds 2 for  $x \geq 200$ ,

$$\begin{aligned} (x - 6.66)^2 \varphi'(x) &= \frac{t^2}{8} L \frac{x - 6.66}{x} - \left( \frac{t^2}{16} L^2 + 0.626 \right) \\ &< \frac{t^2}{8} L - \frac{t^2}{16} L^2 - 0.626 \\ &= -\frac{t^2}{16} L(L - 2) - 0.626 < 0, \end{aligned}$$

so  $\varphi$  is decreasing, and on the window  $\varphi(x) \leq \varphi(X_N(t)) = \Delta(N, t)$ : the per-term factor is at most  $e^{\Delta(N, t)} - 1$ .

*Assembly of  $E_{AB}$ .* By (71)–(72) of [8] and Lemma 5.1 ( $\operatorname{Re} s_* \geq \sigma_1$ ,  $|\gamma| \leq G$ , and  $n^{y - \operatorname{Re} s_* - \operatorname{Re} \kappa} \leq n^{-\sigma_2}$ ),

$$e_B \leq (e^{\Delta(N, t)} - 1) \sum_{n \leq N} b_t(n) n^{-\sigma_1}, \quad e_A \leq G(e^{\Delta(N, t)} - 1) \sum_{n \leq N} b_t(n) n^{-\sigma_2}.$$

Splitting each sum at  $M_{\text{err}}$  and bounding the range  $M_{\text{err}} < n \leq N$  by Lemma 5.2 (legitimate by (28)) gives  $e_A + e_B \leq E_{AB}(N, t, y)$ .

*Bound for  $e_{C,0}$ .* By the corollary (19),  $e_{C,0}$  is bounded by the product of  $(\frac{x}{4\pi})^{-(1+y)/4}$ ,  $\exp(-\frac{t}{16} \log^2 \frac{x}{4\pi})$ ,  $\exp(1.24(3^y + 3^{-y})/(N - 0.125))$ , and  $\exp(\psi(x))$  with  $\psi(x) = (3\sqrt{L^2 + \pi^2/4} + 10.50)/(x - 12)$ . The first two factors are decreasing in  $x$ , the third does not depend on  $x$ , and  $\psi$  is decreasing: with  $Q(L) = 3\sqrt{L^2 + \pi^2/4} + 10.50$  one has  $\frac{d}{dx}Q = \frac{3L}{\sqrt{L^2 + \pi^2/4}} \cdot \frac{1}{x} < \frac{3}{x}$ , whence  $(x - 12)^2 \psi'(x) < 3 \frac{x-12}{x} - Q < 3 - 10.50 < 0$ . Evaluating all four factors at the left edge  $x = X_N(t)$ , where  $\log \frac{x}{4\pi} = L_N(t)$ , gives  $e_{C,0} \leq E_C(N, t, y)$ .

*Decay in  $N$ .* In  $E_{AB}$ : the factor  $e^{\Delta(N, t)} - 1$  is nonincreasing because  $\Delta(N, t) = \varphi(X_N(t))$  with  $\varphi$  decreasing and  $X_N(t)$  increasing in  $N$ ; the heads of  $S_1, S_2$  and the factor  $G$  are nonincreasing by Lemma 5.6; and the caps  $\text{Cap}_t(M_{\text{err}}, N; \sigma_i)$  are nonincreasing by Lemma 5.5 with  $a = M_{\text{err}}$ , whose conditions (31) hold by hypothesis. In  $E_C$ : the factors  $(N^2 - t/16)^{-(1+y)/4}$ ,  $e^{-\frac{t}{16} L_N(t)^2}$  and  $e^{1.24(3^y + 3^{-y})/(N - 0.125)}$  are visibly nonincreasing in  $N$ , and  $e^{\psi(X_N(t))}$  is nonincreasing because  $\psi$  is decreasing and  $X_N(t)$  increases. Products of nonnegative nonincreasing factors are nonincreasing.  $\square$

## 5.6 The cutoff inequalities and the tail theorem

The computational input of the tail argument is the following.

**Proposition 5.10** (Cutoff inequalities). *The cap-validity conditions (28) and the cutoff conditions (31)–(32) hold, uniformly for  $t \in I_t$  and  $y \in [y_0, \sqrt{1 - 2t}]$ , and*

$$\begin{aligned} \sup_{t \in I_t, y \in I_{\text{box}}} D(N_*, t, y) &< 0.999721, & \sup_{t \in I_t, y \in I_{\text{box}}} M_{\text{max}}(N_*, t, y) &< 1.608290, \\ \sup_{t \in I_t, y \in I_{\text{ext}}} (E_{AB} + E_C)(N_*, t, y) &< 1.1672 \times 10^{-8}, \end{aligned}$$

with  $E_{AB}, E_C$  the error majorants of Lemma 5.9. In addition, a direct joint interval evaluation, followed by the reductions of Lemmas 5.8 and 5.9, gives for every  $N \geq N_*$ ,  $t \in I_t$ ,  $y \in [y_0, \sqrt{1 - 2t}]$ , and  $x \in [X_N(t), X_{N+1}(t)]$  satisfies

$$\frac{1 - D(N, t, y)}{M_{\text{max}}(N, t, y)} - (e_A(x, y, t) + e_B(x, y, t) + e_{C,0}(x, y, t)) > 0.00017352 > 0.$$



*Proof.* The primary sealed evidence is [verifiers/verify\\_tail\\_arb.c](#), the transcript checker [verifiers/verify\\_tail\\_arb\\_logs.py](#), and the preserved transcripts [logs/tail\\_arb\\_256.log](#) and [logs/tail\\_arb\\_512.log](#). A derived second-toolchain reconstruction, [dan-reworking/code/prop510/prop510\\_proof.c](#), reproduces every condition and displayed constant below. It reads no stored data; every decisive comparison is a strict ball inequality, which fails closed when enclosures overlap; and it is run at both 256 and 512 bits of precision with identical verdicts.

*How the suprema are captured.* The intervals  $I_t$ ,  $I_{\text{box}}$  and  $I_{\text{ext}}$  enter the computation as single balls whose endpoints are the exact rational interval endpoints; consequently every derived quantity is an enclosure valid simultaneously for all  $(t, y)$  in the corresponding hull, and its directed upper (resp. lower) endpoint bounds the supremum (resp. infimum) over that hull. More precisely, for each certified precision  $p \in \{256, 512\}$  let  $\text{Up}_p(E)$  denote the exact dyadic upper endpoint returned by Arb’s directed upper-endpoint extraction for the  $p$ -bit ball evaluation of an expression  $E$  over  $I_t$ . That run uses

$$\hat{\delta}_p = \text{Up}_p(\delta(N_*, I_t)), \quad \hat{k}_p = \text{Up}_p(k(N_*, I_t)),$$

which satisfy  $\hat{\delta}_p \geq \hat{\delta}$  and  $\hat{k}_p \geq \hat{k}$ . It substitutes these dyadic values for the exact maxima and extracts point lower bounds for  $\sigma_1, \sigma_2$  over each hull. This is conservative throughout: increasing either frozen defect can only lower the exponents, and every subsequent occurrence of  $\sigma_i$  is through  $u^{-\sigma_i}$  or  $E_{t, \sigma_i}(u)$  with  $u \geq 1$ , both nonincreasing in  $\sigma_i$ .

*What is checked.* (1) The cap-validity conditions (28), in the equivalent form  $\sigma_i > \frac{t}{2} \log N_*$ , on both hulls (labels SC1, SC2 in the program’s output). (2) The cutoff conditions (31) for exactly the left endpoints listed there:  $a = \lfloor M/d \rfloor$  for each  $d \in \mathcal{S}$  at  $\sigma_1$  (GN-TR),  $a = M$  at  $\sigma_2$  (GN-AB), and  $a = M_{\text{err}}$  at both  $\sigma_1$  and  $\sigma_2$  (GN-error), each with the  $t$ -factor evaluated at  $\min I_t$  and the  $\sigma$ -factor at its lower bound over  $I_{\text{ext}}$ . (3) The  $A$ -side slope condition (32) (YM). (4) The hypotheses of Lemmas 5.5–5.9:  $\sigma_1, \sigma_2 > 1$  on  $I_{\text{ext}}$ ,  $N_*(1 - \frac{t}{16N_*^2}) > 1$ ,  $X_{N_*}(t) > 200$ ,  $\log(N_*^2 - \frac{t}{16}) > 2$ , and the sign conditions making  $\Delta(N, t)$  and the  $e_{C,0}$  quotient decreasing. (5) The exact rational identities: the mollifier normalization  $\sum_d |\lambda_d| = 9.918746$ ,  $\lambda_1 = 1$ ,  $t_0 + y_0^2/2 = \frac{893927}{5 \cdot 10^6}$ , and the hull containments  $y_0 \in I_{\text{box}}$ ,  $\max I_{\text{box}} \leq \sqrt{1 - 2t}$  for all  $t \in I_t$ , and  $\max I_{\text{ext}} \geq \sqrt{1 - 2t_0}$ , all checked in exact integer arithmetic. (6) The quantity  $D$  is assembled per (27) from an exact Dirichlet convolution  $c_m(t)$  computed by direct double summation over  $(d, k)$ , and the four displayed constants are established as strict ball comparisons against exact rationals:

$$D < 0.999721, \quad M_{\text{max}} < 1.608290, \quad E_{AB} + E_C < 1.1672 \times 10^{-8},$$

$$\frac{1 - D}{M_{\text{max}}} - (E_{AB} + E_C) > 0.00017352.$$

(The program additionally performs two-sided “corridor” checks on each constituent quantity, which expose missing or duplicated terms.) The enclosures obtained at 512 bits include  $D \leq 0.9997209094$ ,  $M_{\text{max}} \leq 1.6082891450$ ,  $E_{AB} + E_C \leq 1.16716035 \times 10^{-8}$ , and margin  $\geq 0.0001735209$ . Both runs (256 and 512 bits) pass all 37 checks; the reports are recorded as [dan-reworking/code/prop510/runs/prop510\\_report\\_256.txt](#) and [dan-reworking/code/prop510/runs/prop510\\_report\\_512.txt](#), and each run completes in under a second. The reconstruction and reports are unsealed and corroborative; the sealed verifier, checker, and transcripts are the primary candidate evidence.  $\square$

**Claim 5.11** (Candidate tail conclusion). *For every  $t \in I_t$ ,  $y \in [y_0, \sqrt{1-2t}]$  and  $x \geq X_{N^*}(t)$ ,*

$$|f_t(x + iy)| > e_A + e_B + e_{C,0},$$

*and consequently  $H_t(x + iy) \neq 0$  on this region.*

*Proof.* Let  $N$  be the window index of  $x$ , so  $N \geq N^*$ . By Lemma 5.3, Lemma 5.8 and Proposition 5.10,

$$|f_t(x + iy)| \geq \frac{1 - D(N, t, y)}{M_{\max}(N, t, y)} > e_A + e_B + e_{C,0},$$

where the strict final comparison is precisely the direct interval inequality in Proposition 5.10, after applying both clauses of Lemma 5.8 and Lemma 5.9. Here  $[y_0, \sqrt{1-2t}] \subseteq I_{\text{ext}}$ . The point  $(x, y, t)$  lies in the region (10), so [8, Corollary 1.4] gives  $H_t(x + iy) \neq 0$ .  $\square$

## 6 Hypothesis (iii): the barrier

### 6.1 Statement, and reduction to a closed rectangle

Throughout this section let

$$R = [X, X + 1] + i \left[ \frac{1809}{10000}, 1 \right].$$

The goal of the section is the following candidate conclusion, whose argument is assembled in Section 6.5.

**Claim 6.1** (Candidate barrier conclusion).  *$H_t(z) \neq 0$  for all  $z \in R$  and  $0 \leq t \leq t_0$ .*

Claim 6.1 implies hypothesis (iii) of Theorem 2.1: the curved barrier region of that hypothesis satisfies, horizontally,  $X \leq x \leq X + \sqrt{1 - y_0^2} \leq X + 1$ ; and vertically,  $y \geq \sqrt{y_0^2 + 2(t_0 - t)} \geq y_0 > \frac{1809}{10000}$  (indeed  $y_0^2 - (\frac{1809}{10000})^2 = \frac{234599}{10^8} > 0$ ) and  $y \leq \sqrt{1-2t} \leq 1$ ; hence it is contained in  $R \times [0, t_0]$ .

### 6.2 Uniform effective error on the box, including $t = 0$

**Proposition 6.2** (Uniform error bound on the barrier box). *For all  $x \in [X, X + 1]$  and  $0 \leq t \leq t_0$ , the window index satisfies  $N = \lfloor \sqrt{\frac{x}{4\pi}} + \frac{t}{16} \rfloor = 690988$  and  $N^2 < x/(4\pi)$ . Moreover, for all  $z \in R$  and  $0 < t \leq t_0$ , the error terms of Theorem 1.3, bounded with the constants of Section 2.3, satisfy*

$$e_A + e_B + e_{C,0} < 0.000356523011600040 < 0.00125. \quad (33)$$

*Proof.* The primary sealed evidence is [barrier/src/verify\\_uniform\\_error\\_01787854.c](#) and the preserved certificate [barrier/certificates/uniform\\_error\\_256.log](#). The derived reconstruction [independent/prop62/prop62\\_proof.c](#), sealed and executed by the quick verifier at snapshot [034a37acce703e128e63e2f36520b0c8af6c1c76](#), encloses every quantity below in 256-bit ball arithmetic starting from the exact inputs  $X$ ,  $t_0 = \frac{129}{800}$ ,  $y_{\min} = \frac{1809}{10000}$  and  $N = 690988$ ; it reads no stored data, and every acceptance condition is a strict ball inequality, which fails closed.

*Window-index constancy.* The quantity  $q(x, t) = \frac{x}{4\pi} + \frac{t}{16}$  is strictly increasing in each of  $x$  and  $t$ , and  $N = \lfloor \sqrt{q} \rfloor$  is nondecreasing in  $q$ ; so it suffices to evaluate  $N$  at the two extreme corners  $(X, 0)$  and  $(X + 1, t_0)$  of the box. At each corner the program produces an enclosure of  $\lfloor \sqrt{q} \rfloor$  that collapses to an exact integer (a straddling enclosure would fail closed) and checks that both integers equal 690988. It further checks  $N^2 < X/(4\pi) \leq x/(4\pi)$ , and the equivalent inequality  $2 \log N < L_{\min} := \log \frac{X}{4\pi}$ ; write also  $L_{\max} := \log \frac{X+1}{4\pi}$ .

*Error bound.* With  $N$  constant, for  $0 < t \leq t_0$  the error terms are bounded by equation (23) of [8] and by (19); the program encloses the following majorants, each uniform on the box because every  $t$ - and  $y$ -dependent factor is evaluated at its worst endpoint: (i) by (17), using  $(\cdot)_+ \leq 1 + 8/X^2$  and  $y \geq y_{\min}$ ,  $\operatorname{Re} s_* \geq a + \frac{t}{4} L_{\min} - \delta$  with  $a = \frac{1+y_{\min}}{2}$  and  $\delta = \frac{t_0(1+8/X^2)}{2X^2}$ ; since  $\log n \leq \log N < \frac{1}{2} L_{\min}$ , this gives  $b_t(n) n^{-\operatorname{Re} s_*} \leq n^{-a_{\text{eff}}}$  with  $a_{\text{eff}} = a - \delta$ , uniformly in  $t \in [0, t_0]$  (the  $t$ -dependent parts cancel against  $\frac{t}{4} \log^2 n$ ; checked:  $0 < a_{\text{eff}} < 1$ ); (ii) hence, by comparison with  $\int_1^N u^{-a_{\text{eff}}} du$ ,  $\sum_{n=1}^N b_t(n) n^{-\operatorname{Re} s_*} \leq 1 + \frac{N^{1-a_{\text{eff}}}-1}{1-a_{\text{eff}}}$ ; (iii) since  $n^2 \leq N^2 < x/(4\pi)$ , the bounds (16) and (18) give  $|\gamma| n^y \leq e^{0.02y} \left(\frac{4\pi n^2}{x}\right)^{y/2} \leq e^{0.02}$  and  $N^{|\kappa|} \leq \exp(\frac{t_0 \log N}{2(X-6)})$ , so the prefactor of (23) is at most  $1 + e^{0.02} \exp(\frac{t_0 \log N}{2(X-6)})$ ; (iv) since  $0 < \log \frac{x}{4\pi n^2} \leq L_{\max}$ , the last factor of (23) is at most  $\exp(\frac{(t_0^2/16)L_{\max}^2 + 0.626}{X-6.66}) - 1$ ; (v) in (19), the prefactor is maximized at  $(x, y) = (X, y_{\min})$ , the factor  $e^{-\frac{t}{16} \log^2 \frac{x}{4\pi}}$  is at most 1,  $3^y + 3^{-y} \leq 3 + \frac{1}{3} = \frac{10}{3}$  (increasing in  $y$ , and  $y \leq 1$ ), and the final quotient is bounded by maximizing its numerator ( $L \leq L_{\max}$ ) and minimizing its denominator ( $x \geq X$ ) separately. The program multiplies and sums these enclosures — yielding  $e_A + e_B \leq 3.64 \times 10^{-10}$  and  $e_{C,0} \leq 0.000356522648$  — and establishes the two strict inequalities of (33), the sharp constant being compared as the exact rational  $356523011600040 \times 10^{-18}$ . The run completes in under a second; its report is retained, unsealed, at [dan-reworking/code/prop62/runs/prop62\\_report.txt](#). The reconstruction cross-checks the sharp displayed digit but is not the primary candidate evidence.  $\square$

The constancy of  $N$  makes  $f_t$  a single analytic expression on  $R$ . The round number  $0.00125 = \frac{1}{800}$  is the *approximation allowance* used by the prism criterion below; (33) says it is safely conservative.

Theorem 1.3 is stated in [8] for  $t > 0$ ; the following lemma extends its conclusion to the closed endpoint  $t = 0$  on the compact box.

**Lemma 6.3** (Extension to  $t = 0$ ). *For all  $z \in R$  and all  $0 \leq t \leq t_0$  (including  $t = 0$ ),  $B_t(z) \neq 0$  and*

$$|g_t(z) - f_t(z)| \leq 0.00125,$$

where  $g_t = H_t/B_t$  and  $f_t$  is the sum (12) with the constant window index  $N = 690988$ .

*Proof.* For  $0 < t \leq t_0$  this is Theorem 1.3 together with Proposition 6.2. For the endpoint, fix  $z \in R$  and argue by a limit  $t_j \downarrow 0$ : (1) by dominated convergence in (1) (majorant  $e^{t_0 u^2 + u} |\Phi(u)|$ , integrable by the super-exponential decay of  $\Phi$ ),  $H_t(z)$  is jointly continuous down to  $t = 0$  on  $R$ ; (2)  $B_t^{\pm 1}$  is continuous and  $B_t \neq 0$  on  $R \times [0, t_0]$ , because the argument  $\frac{1+y-ix}{2}$  of  $M_0$  stays away from the branch cut (Section 2.2) and the deforming exponential never vanishes; (3) since the window index is constant,  $f_t$  is a fixed finite sum of functions continuous in  $t$ ; (4) the error majorant is continuous in  $t$  and bounded by 0.00125 on  $(0, t_0]$  by (33). Passing to the limit in  $|g_{t_j} - f_{t_j}| \leq 0.00125$  gives the claim at  $t = 0$ .  $\square$

### 6.3 Uniform derivative bounds on the box

The moving-mesh argument below needs uniform bounds for  $|\partial_z f_t|$  and  $|\partial_t f_t|$  on  $R$ . The pointwise input is the following result; recall that by Proposition 6.2 the window index is constantly  $N = 690988$  on the box, so that  $f_t$  is the fixed finite sum (12) with this  $N$ , jointly smooth in  $(z, t)$ , and the lemma applies to it verbatim.

**Lemma 6.4** ([8, Lemma 8.4]). *In the region (10), and away from the jump discontinuities of  $N$ ,*

$$\begin{aligned} \left| \frac{\partial f_t}{\partial z} \right| &\leq \sum_{n=1}^N \frac{b_t(n)}{n^{\operatorname{Re} s_*}} \left( \frac{\log n}{2} + \frac{t \log n}{4(x-6)} \right) \\ &\quad + |\gamma| N^{|\kappa|} \sum_{n=1}^N \frac{b_t(n) n^y}{n^{\operatorname{Re} s_*}} \left( \frac{t \log n}{4(x-6)} + \left( \log \frac{|1+y+ix|}{4\pi} + \pi + \frac{3}{x} \right) \left( \frac{1}{2} + \frac{t}{4(x-6)} \right) \right), \end{aligned}$$

and

$$\begin{aligned} \left| \frac{\partial f_t}{\partial t} \right| &\leq \sum_{n=1}^N \frac{b_t(n)}{n^{\operatorname{Re} s_*}} \left( \frac{1}{4} \log n \log \frac{x}{4\pi n} + \frac{\pi}{8} \log n + \frac{2 \log n}{x-6} \right) \\ &\quad + |\gamma| N^{|\kappa|} \sum_{n=1}^N \frac{b_t(n) n^y}{n^{\operatorname{Re} s_*}} \left( \frac{1}{4} \log n \log \frac{x}{4\pi n} + \frac{\pi}{8} \log n + \frac{2 \log n}{x-6} \right. \\ &\quad \left. + \frac{1}{4} \left( \frac{\pi}{2} + \frac{8}{x-6} \right) \left( \log \frac{x}{4\pi} + \frac{8}{x-6} \right) \right). \end{aligned}$$

To convert these pointwise bounds into uniform ones we introduce the following frozen quantities. Write  $y_{\min} = \frac{1809}{10000}$  (the bottom of  $R$ ),  $q(x) = \frac{x}{4\pi}$ , and, as in (17),

$$h(x, y) = 1 - 3y + \frac{4y(1+y)}{x^2}, \quad c(x, y) = \frac{1}{4} \log q(x) - \frac{h(x, y)_+}{2x^2}, \quad c_0 = c(X, y_{\min}),$$

so that (17) reads  $\operatorname{Re} s_* \geq \frac{1+y}{2} + t c(x, y)$ . Define further

$$\begin{aligned} a(t) &= e^{0.02 y_{\min}} q(X)^{-y_{\min}/2} N^{t y_{\min}/(2(X-6))}, \\ P_t(u) &= u^{-\frac{1+y_{\min}}{2} + t(\frac{1}{4} \log u - c_0)}, \\ \theta(t) &= \frac{1}{2} + \frac{t}{4(X-6)}, \\ \Lambda_z(t) &= \left( \log \frac{|2+i(X+1)|}{4\pi} + \pi + \frac{3}{X} \right) \theta(t), \\ \Lambda_t &= \frac{1}{4} \left( \frac{\pi}{2} + \frac{8}{X-6} \right) \left( \log \frac{X+1}{4\pi} + \frac{8}{X-6} \right), \\ C_2 &= \frac{1}{4} \log \frac{X+1}{4\pi} + \frac{\pi}{8} + \frac{2}{X-6}, \end{aligned}$$

and the two summand majorants

$$\begin{aligned} F_z(u; t) &= P_t(u) \left[ \theta(t) \log u + a(t) u^{y_{\min}} \left( \frac{t \log u}{4(X-6)} + \Lambda_z(t) \right) \right], \\ F_t(u; t) &= P_t(u) \left[ (1 + a(t) u^{y_{\min}}) \log u \left( C_2 - \frac{\log u}{4} \right) + a(t) u^{y_{\min}} \Lambda_t \right], \end{aligned}$$

and finally

$$D_z(t) = \sum_{n=1}^{16} F_z(n; t) + \int_{16}^N F_z(u; t) du, \quad D_t(t) = \sum_{n=1}^{16} F_t(n; t) + \int_{16}^N F_t(u; t) du.$$

Two tight inequalities at the exact barrier parameters are needed below. The first,  $N^2 < q(X)$ , is part of Proposition 6.2. The second we record as a standalone statement.

**Proposition 6.5** (Tail-exponent inequality). *At  $X = 6000000185827$ ,  $N = 690988$ ,  $y_{\min} = \frac{1809}{10000}$ ,*

$$\frac{1}{4} \log \frac{q(X)}{N^2} > \frac{h(X, y_{\min})_+}{2X^2}.$$

*Proof.* The primary sealed evidence is the run-time gate in [barrier/src/TloopSinglemat\\_closed\\_cert.c](#), the preserved transcript [barrier/certificates/barrier\\_target\\_closed.log](#), and its fail-closed checker [verifiers/verify\\_barrier\\_binding.py](#). The derived unsealed reconstruction [dan-reworking/code/prop65/prop65\\_proof.c](#) encloses both sides in 256-bit ball arithmetic from the exact rational parameters and establishes the displayed strict inequality as a strict ball comparison, which fails closed. The computed enclosures are

$$\begin{aligned} \frac{1}{4} \log \frac{q(X)}{N^2} &= 2.2405813008 \dots \times 10^{-7} \pm 10^{-31}, \\ \frac{h(X, y_{\min})_+}{2X^2} &= 6.3513884954 \dots \times 10^{-27} \pm 10^{-51}. \end{aligned}$$

the inequality is tight only in the sense that its left side is small — it is positive precisely because  $N^2 < q(X)$ , an inequality of relative size approximately  $9 \times 10^{-7}$  — while the right side is smaller by twenty orders of magnitude. The run completes in milliseconds; its output is recorded as [dan-reworking/code/prop65/runs/prop65\\_report.txt](#). The same inequality is re-checked at run time by the sealed program of Proposition 6.12 (like the conditions (C2), (C4) and (C7) of Proposition 6.6).  $\square$

The construction consumes the following list of strict inequalities at the exact barrier parameters.

**Proposition 6.6** (Derivative-bound conditions). *At  $X = 6000000185827$ ,  $N = 690988$ ,  $y_{\min} = \frac{1809}{10000}$ ,  $t_0 = \frac{129}{800}$ :*

- (C1)  $X^2 > 8$  and  $\frac{1}{4(X+1)} - \frac{2}{X^3} > 0$ ;
- (C2)  $0.02 - \frac{1}{2} \log q(X) + \frac{t_0 \log N}{2(X-6)} + \frac{1}{2} \log N < 0$ ;
- (C3)  $2 \log N < \log q(X)$ ;
- (C4)  $\frac{12}{X^2} < 3$ ;
- (C5)  $c_0 > \frac{1}{2} \log N$ ;
- (C6)  $\frac{1-y_{\min}}{2} > \frac{1}{\log 16}$ ;
- (C7)  $C_2 > \frac{1}{4} \log N$ .

*Proof.* (C1), (C4) and (C6) are immediate by inspection ( $X \approx 6 \times 10^{12}$ ;  $\frac{1}{\log 16} = 0.3607 \dots < 0.40955 = \frac{1-y_{\min}}{2}$ ). For (C2), rewrite the left side as  $0.02 + \frac{1}{2} \log \frac{4\pi N}{X} + \frac{t_0 \log N}{2(X-6)}$ : here  $\frac{4\pi N}{X} < \frac{8.8 \times 10^6}{6 \times 10^{12}} < 2 \times 10^{-6} < e^{-13}$ , so the middle term is less than  $-\frac{13}{2}$ , while the last term is less than  $10^{-12}$ ; hence the left side is less than  $0.02 - 6.5 + 10^{-12} < 0$ . (C7) follows from (C3):  $C_2 - \frac{1}{4} \log N \geq \frac{1}{4} \log q(X) - \frac{1}{4} \log N > \frac{1}{4} \log N > 0$ .

The remaining two conditions are exactly the statements established earlier: (C3) is the inequality  $N^2 < q(X)$  of Proposition 6.2, and (C5) is Proposition 6.5, since  $c_0 - \frac{1}{2} \log N = \frac{1}{4} \log \frac{q(X)}{N^2} - \frac{h(X, y_{\min})_+}{2X^2}$ . (The program of Proposition 6.12 also re-checks (C2), (C4) and (C7) at run time; by the above, those checks are mathematically redundant, though harmless.)  $\square$

**Lemma 6.7** (Pointwise domination on the box). *Assume (C1), (C2), (C4). Then for all  $x \in [X, X+1]$ ,  $y \in [y_{\min}, 1]$ ,  $0 \leq t \leq t_0$  and  $1 \leq n \leq N$ , the  $n$ -th summand of the first (resp. second) bound of Lemma 6.4 is at most  $F_z(n; t)$  (resp.  $F_t(n; t)$ ). Consequently, for  $0 \leq t \leq t_0$ ,*

$$\sup_{z \in R} \left| \frac{\partial f_t}{\partial z}(z) \right| \leq \sum_{n=1}^N F_z(n; t), \quad \sup_{z \in R} \left| \frac{\partial f_t}{\partial t}(z) \right| \leq \sum_{n=1}^N F_t(n; t).$$

*Proof. Step 1:  $c$  is minimized at  $(X, y_{\min})$ .* In  $y$ :  $\partial_y h = -3 + \frac{4(1+2y)}{x^2} \leq -3 + \frac{12}{X^2} < 0$  by (C4), so  $h$ , hence  $h_+$ , is nonincreasing in  $y$  and  $c$  is nondecreasing in  $y$ . In  $x$ : where  $h > 0$ ,

$$\left| \partial_x \frac{h}{2x^2} \right| = \left| -\frac{4y(1+y)}{x^5} - \frac{h}{x^3} \right| \leq \frac{8}{x^5} + \frac{h_+}{x^3} \leq \frac{2}{x^3},$$

using  $4y(1+y) \leq 8$ ,  $8/x^2 \leq 1$  and  $h_+ \leq 1$  (both from  $X^2 > 8$ ; for  $y > 0$ ,  $h \leq 1$  reduces to  $4(1+y) \leq 3x^2$ , and at  $y = 0$ ,  $h = 1$ ). Where  $h < 0$  the positive part is locally constant, and  $h_+/(2x^2)$  is locally Lipschitz across the kink with the same bound. Hence for  $x \in [X, X+1]$ ,  $\partial_x c \geq \frac{1}{4x} - \frac{2}{X^3} \geq \frac{1}{4(X+1)} - \frac{2}{X^3} > 0$  almost everywhere by (C1), so  $c$  is increasing in  $x$ . Therefore  $c(x, y) \geq c(X, y_{\min}) = c_0$  on the box.

*Step 2: the B-side.* By (17) and Step 1,  $\operatorname{Re} s_* \geq \frac{1+y}{2} + t c(x, y) \geq \frac{1+y_{\min}}{2} + t c_0$ , so

$$b_t(n) n^{-\operatorname{Re} s_*} \leq n^{\frac{t}{4} \log n - \frac{1+y_{\min}}{2} - t c_0} = P_t(n),$$

while the B-side brackets obey  $\log n(\frac{1}{2} + \frac{t}{4(x-6)}) \leq \theta(t) \log n$  and

$$\log n \left( \frac{1}{4} \log \frac{x}{4\pi n} + \frac{\pi}{8} + \frac{2}{x-6} \right) = \log n \left( \frac{1}{4} \log q(x) - \frac{\log n}{4} + \frac{\pi}{8} + \frac{2}{x-6} \right) \leq \log n \left( C_2 - \frac{\log n}{4} \right),$$

since  $q$  is increasing and  $\frac{1}{x-6}$  decreasing on  $[X, X+1]$ .

*Step 3: the A-core.* By (16)–(18),  $|\gamma| N^{|\kappa|} n^y b_t(n) n^{-\operatorname{Re} s_*} \leq b_t(n) \mathcal{A}(x, y, t; n)$  with

$$\mathcal{A}(x, y, t; n) = e^{0.02y} q(x)^{-y/2} N^{\frac{ty}{2(x-6)}} n^{y - \frac{1+y}{2} - t c(x, y)}.$$

Using  $\partial_y c \geq 0$  (Step 1) and  $\log n \geq 0$ ,

$$\begin{aligned} \partial_y \log \mathcal{A} &= 0.02 - \frac{1}{2} \log q(x) + \frac{t \log N}{2(x-6)} + \frac{\log n}{2} - t (\partial_y c) \log n \\ &\leq 0.02 - \frac{1}{2} \log q(X) + \frac{t_0 \log N}{2(X-6)} + \frac{\log N}{2}, \end{aligned}$$

which is negative by (C2); so  $\mathcal{A}$  is maximized at  $y = y_{\min}$ . At  $y = y_{\min}$ , freezing  $x$  conservatively ( $q(x)^{-y_{\min}/2} \leq q(X)^{-y_{\min}/2}$ ,  $N^{\frac{ty_{\min}}{2(x-6)}} \leq N^{\frac{ty_{\min}}{2(X-6)}}$ ,  $c(x, y_{\min}) \geq c_0$ ) gives  $b_t(n) \mathcal{A} \leq a(t) n^{y_{\min}} P_t(n)$ .

*Step 4: the A-side brackets.* On  $[X, X+1] \times [y_{\min}, 1]$ , we have

$$\frac{t \log n}{4(x-6)} \leq \frac{t \log n}{4(X-6)},$$

$$\left( \log \frac{|1+y+ix|}{4\pi} + \pi + \frac{3}{x} \right) \left( \frac{1}{2} + \frac{t}{4(x-6)} \right) \leq \left( \log \frac{|2+i(X+1)|}{4\pi} + \pi + \frac{3}{X} \right) \theta(t) = \Lambda_z(t),$$

since  $|1+y+ix| \leq |2+i(X+1)|$  (both coordinates increase) and  $\frac{3}{x} \leq \frac{3}{X}$ ; and

$$\frac{1}{4} \left( \frac{\pi}{2} + \frac{8}{x-6} \right) \left( \log q(x) + \frac{8}{x-6} \right) \leq \Lambda_t,$$

each factor being bounded by its value at the appropriate endpoint.

Combining Steps 2–4 termwise bounds the two summands of Lemma 6.4 by  $F_z(n; t)$  and  $F_t(n; t)$ . For  $0 < t \leq t_0$  the displayed conclusions are then Lemma 6.4; at  $t = 0$  they follow by continuity, as both sides of each inequality are continuous in  $t$  on the compact box ( $f_t$  being the fixed finite sum).  $\square$

**Lemma 6.8** (Head-and-tail evaluation). *Assume (C5), (C6), (C7). Then for each fixed  $t \in [0, t_0]$  the functions  $u \mapsto F_z(u; t)$  and  $u \mapsto F_t(u; t)$  are positive and nonincreasing on  $[16, N]$ , and consequently*

$$\sum_{n=1}^N F_z(n; t) \leq D_z(t), \quad \sum_{n=1}^N F_t(n; t) \leq D_t(t).$$

*Proof.* Write  $r = \log u \in [\log 16, \log N]$ . The logarithmic derivative of  $P_t$  with respect to  $r$  is  $p(r) = -\frac{1+y_{\min}}{2} + t(\frac{r}{2} - c_0) \leq -\frac{1+y_{\min}}{2}$ , since  $t \geq 0$  and  $\frac{r}{2} \leq \frac{1}{2} \log N \leq c_0$  by (C5). Each of  $F_z, F_t$  is a nonnegative linear combination (with  $u$ -independent coefficients  $\theta(t)$ ,  $\frac{t}{4(X-6)}$ ,  $a(t)$ ,  $\Lambda_z(t)$ ,  $\Lambda_t$ , all  $\geq 0$ ) of the five component functions

$$P_t(u)r, \quad P_t(u)u^{y_{\min}}r, \quad P_t(u)u^{y_{\min}}, \quad P_t(u)B(r), \quad P_t(u)u^{y_{\min}}B(r),$$

with  $B(r) = r(C_2 - \frac{r}{4})$ , which is positive on  $[\log 16, \log N]$  by (C7). Their logarithmic  $r$ -derivatives are, respectively, at most

$$p + \frac{1}{r}, \quad p + y_{\min} + \frac{1}{r}, \quad p + y_{\min}, \quad p + \frac{B'}{B}, \quad p + y_{\min} + \frac{B'}{B},$$

and  $\frac{B'}{B} = \frac{1}{r} - \frac{1}{4(C_2 - r/4)} \leq \frac{1}{r}$ . Hence every component has logarithmic derivative at most

$$p + y_{\min} + \frac{1}{\log 16} \leq -\frac{1+y_{\min}}{2} + y_{\min} + \frac{1}{\log 16} = -\frac{1-y_{\min}}{2} + \frac{1}{\log 16} < 0$$

by (C6), so each component, and therefore each of  $F_z, F_t$ , is positive and decreasing on  $[16, N]$ . Finally, for decreasing positive  $F$ ,  $\sum_{n=17}^N F(n) \leq \int_{16}^N F(u) du$  (compare each term with the integral over  $[n-1, n]$ ), which gives the displayed bounds.  $\square$



**Lemma 6.9** (Uniform derivative bounds). *Assume (C1)–(C7). Then for every  $t \in [0, t_0]$ ,*

$$\sup_{z \in R} \left| \frac{\partial f_t}{\partial z}(z) \right| \leq D_z(t), \quad \sup_{z \in R} \left| \frac{\partial f_t}{\partial t}(z) \right| \leq D_t(t),$$

*and hence, for every closed interval  $[t_i, t_{i+1}] \subseteq [0, t_0]$ , the relation*

$$\sup_{z \in R, \tau \in [t_i, t_{i+1}]} |\partial_\tau f_\tau(z)| \leq \sup_{\tau \in [t_i, t_{i+1}]} D_t(\tau)$$

*holds.*

*Proof.* Combine Lemmas 6.7 and 6.8. □

**Remark 6.10.** The program of Proposition 6.12 evaluates  $D_z(t)$  and  $D_t(t)$  with  $t$  an interval enclosure — for the prism bound, an enclosure of the whole interval  $[t_i, t_{i+1}]$ , which by inclusion isotonicity encloses  $\sup_\tau D_t(\tau)$  — and computes the integrals  $\int_{16}^N F(u) du$  by Arb’s rigorous Gauss–Legendre integrator [2]. That algorithm requires the integrand callback to supply enclosures that are holomorphic on the complex balls it queries; the integrands above extend holomorphically in  $u$  away from the branch cut of the logarithm, and the program uses the analytic logarithm and power operations, which return non-finite enclosures (failing the run) if a queried ball touches the cut. Two points here require care relative to the Polymath-era barrier code from which the program descends: the discrete sums must not be bounded by the expression  $F(1) + \int_1^N F$  (the summand is not monotone near  $n = 1$ ; Lemma 6.8 gives monotonicity only from  $n = 16$  on), whence the head-plus-integral form of  $D_z, D_t$ ; and the integrand callbacks must not project the integration variable to its real part, which would void the integrator’s error analysis.

## 6.4 The prism criterion

The time interval  $[0, t_0]$  will be covered by finitely many closed “prisms”  $R \times [t_i, t_{i+1}]$ . The following lemma reduces nonvanishing of  $H$  on a prism to finitely many verifiable conditions at its initial time  $t_i$ .

**Lemma 6.11** (Prism criterion). *Let  $0 \leq t_i < t_{i+1} \leq t_0$ , and let  $z_1, \dots, z_m$  (cyclically ordered,  $z_{m+1} = z_1$ ) be a mesh on  $\partial R$  with consecutive spacing at most  $h$ . Let  $D_z \geq \sup_{z \in R} |\partial_z f_{t_i}(z)|$  and  $D_t \geq \sup_{z \in R, \tau \in [t_i, t_{i+1}]} |\partial_\tau f_\tau(z)|$  (as supplied by Lemma 6.9), and suppose:*

(a)  $|f_{t_i}(z_j)| \geq M_i$  for all  $j$ , where

$$M_i > \frac{D_z h}{2} + D_t (t_{i+1} - t_i) + 0.00125; \tag{34}$$

(b) *using the clockwise convention implemented by the certificate, each argument increment  $\arg(f_{t_i}(z_j)/f_{t_i}(z_{j+1}))$  lies strictly in  $(-\pi, \pi)$ , and their sum gives winding number 0 for the polygon  $(f_{t_i}(z_j))_j$  about 0.*

*Then  $H_\tau(z) \neq 0$  for all  $z \in R$  and  $\tau \in [t_i, t_{i+1}]$ .*



*Proof.* On each half of a mesh subedge, both the true image curve  $F(s) = f_{t_i}(\text{subedge}(s))$  and its endpoint chord lie in the closed disk of radius  $D_z h/2$  about the nearer endpoint value (this is the factor  $\frac{1}{2}$  in (34)); each disk is convex, so the straight-line homotopy from the true boundary curve to the mesh polygon stays inside these disks. Moving to time  $\tau \in [t_i, t_{i+1}]$  displaces values by at most  $D_t(\tau - t_i)$ , and passing from  $f_\tau$  to  $g_\tau = H_\tau/B_\tau$  by at most 0.00125 (Lemma 6.3). Condition (34) says every resulting disk, centered at a mesh value of modulus  $\geq M_i$ , excludes the origin; hence the entire three-stage homotopy (polygon  $\rightarrow$  true  $f_{t_i}$ -boundary image  $\rightarrow f_\tau(\partial R) \rightarrow g_\tau(\partial R)$ ) avoids zero. In particular each chord avoids the origin, so its argument variation in the implemented clockwise orientation is the principal argument of the stated endpoint ratio. The winding number of the polygon about the origin is a well-defined integer; by (b) it is 0, and the zero-avoiding homotopy gives winding number 0 for  $g_\tau(\partial R)$  as well, for every  $\tau \in [t_i, t_{i+1}]$ . Since  $B_\tau \neq 0$  and  $B_\tau$  is holomorphic on a neighborhood of  $R$  (Lemma 6.3 and Section 2.2),  $g_\tau$  is holomorphic there and has the same zeros as  $H_\tau$ , and  $g_\tau$  is zero-free on  $\partial R$ ; the argument principle then gives  $H_\tau \neq 0$  on  $R$ .  $\square$

## 6.5 The prism decomposition

The computational input of the barrier argument is the following.

**Proposition 6.12** (Prism decomposition). *There exist times  $0 = \tau_0 < \tau_1 < \dots < \tau_{883}$  with  $\tau_{882} < t_0 \leq \tau_{883}$  such that for each  $0 \leq i < 883$ , the hypotheses (a)–(b) of Lemma 6.11 hold for the prism  $R \times [\tau_i, \min(\tau_{i+1}, t_0)]$ , with margin in (34) at least 0.5198498946 in every prism.*

*Proof.* The primary sealed evidence comprises `barrier/src/TloopSinglemat_closed_cert.c`, `barrier/src/StoredSumSinglemat_interval.c`, `barrier/src/StoredSumTaylorTail_cert.c`, the preserved transcripts `barrier/certificates/barrier_target_closed.log`, `barrier/certificates/storedsum_taylor_tail.log`, and `barrier/certificates/store_dsum_provenance.log`, and the fail-closed checker `verifiers/verify_barrier_binding.py`. The derived unsealed reconstruction `dan-reworking/code/prop612/prop612_proof.c` (portions of which descend from the Polymath 15 codebase) reads no stored data, and every decisive comparison is a strict Arb ball comparison, which fails closed. It runs in three phases.

(1) *Domain hypotheses and truncation.* The program first establishes, at 192-bit precision: the uniform expansion-parameter bound  $|w| \leq 0.6480 < 0.66$  on the whole slab (via the shift bounds  $|\text{Re}(\log n_0 - \alpha)| < 0.694$  and  $|\text{Im}(\log n_0 - \alpha)| < 0.7855$  for both  $\alpha$ -arguments used below, where  $n_0 = N/2$ ); the bound  $0.0882 < 0.089$  for the modulus of the  $\gamma$ -type factor, both for its exact expression (on a  $256 \times 128$  subdivision of the box) and for the implemented majorant; and, by a rigorous direct sum over  $n \leq N$  with geometric tail factors, the complete two-variable 62-term Taylor truncation bound  $1.9543 \times 10^{-22} < 10^{-20}$  — the allowance that is attached as an error ball to every boundary mesh value in phase (3).

(2) *Coefficient generation.* The  $62 \times 62$  complex matrix

$$\text{finalmat}[e][k] = \sum_{n=1}^N n^{\frac{i(X+\frac{1}{2})-1}{2}} \frac{(\log \frac{n}{n_0})^e}{e!} \frac{(\frac{1}{4} \log^2 \frac{n}{n_0})^k}{k!}$$

is computed in memory by direct summation, each entry a genuine ball enclosure. A direct calculation with the two Taylor directions shows that the polynomial evaluation of phase (3) applied to this matrix reproduces, exactly up to the truncation bound of phase (1), the two

sums of (12) at the evaluation point, with the heat factor  $b_t(n)$  emerging from the algebra of the two Taylor directions.

(3) *The adaptive prism loop.* Before the loop the program re-checks the conditions (C1)–(C7) of Proposition 6.6 at run time. Then, starting at  $\tau_0 = 0$ : at each prism start  $\tau_i$  it evaluates the derivative bounds  $D_z(\tau_i)$  and (for the candidate prism, with the time parameter interval-valued over the entire closed prism)  $D_t$ , by the head-plus-integral quadratures of Section 6.3, rounding both up to exact integers; meshes  $\partial R$  with  $4\lceil D_z \rceil - 4$  points (spacing at most  $h = 1/(\lceil D_z \rceil - 1)$ ); requires every mesh enclosure of  $f_{\tau_i}$  to exclude zero, every argument increment to lie strictly in  $(-\pi, \pi)$ , and the per-rectangle winding enclosure to lie strictly in  $(-\frac{1}{4}, \frac{1}{4})$ , establishing the integer winding number 0; chooses  $\tau_{i+1}$  adaptively at half the established slack of (34), as an exact dyadic number; and finally re-checks the complete gate  $\min_j |f_{\tau_i}(z_j)| > \frac{D_z h}{2} + D_t(\tau_{i+1} - \tau_i) + 0.00125$  as a strict ball inequality. The loop ends only when the prisms cover  $[0, t_0]$  (adjacent seams are identical exact dyadic numbers, the first time is exactly 0, and a directed comparison checks that the final exact dyadic endpoint satisfies  $\tau_{883} \geq \frac{129}{800} = t_0$ ), and the aggregate winding enclosure must lie strictly in  $(-\frac{1}{4}, \frac{1}{4})$ . (The final prism may extend slightly beyond  $t_0$ ; restricting it to its intersection with  $[0, t_0]$  only shrinks the displacement terms in (34), so the conclusion for the full final prism covers the truncated one, as required by the statement.)

The run establishes 883 prisms, with aggregate winding enclosure  $[\pm 1.18 \times 10^{-14}]$  and worst margin  $0.51984989461387254365 \pm 2 \times 10^{-20}$ , checked against the exact rational threshold  $5198498946 \times 10^{-10}$ . The run completes in under eight minutes; its output is recorded as [dan-reworking/code/prop612/runs/prop612\\_report.txt](#). The reconstruction is a derived corroborative run; the sealed source and certificate chain is the authority for the candidate assertion.  $\square$

*Candidate proof of Claim 6.1.* By Proposition 6.12 and Lemma 6.11,  $H_\tau(z) \neq 0$  for  $z \in R$  and  $\tau \in [\tau_i, \min(\tau_{i+1}, t_0)]$ , for each  $i$ . The union of these intervals is  $[0, t_0]$ .  $\square$

## 7 Assembly of the candidate proof

*Candidate proof of Claim 1.1.* The parameters (4) satisfy  $0 < t_0 < \frac{1}{2}$  and  $0 < y_0^2 < 1 - 2t_0$ . Hypothesis (i) of Theorem 2.1 holds by Section 3, since  $X/2 < T_{PT}$ . Hypothesis (ii) holds by combining Proposition 4.11 (nonvanishing of  $H_{t_0}$  for  $x_* \leq x < x_{3840001}$ ,  $y_0 \leq y \leq \sqrt{1 - 2t_0}$ ) with Claim 5.11 applied at  $t = t_0 \in I_t$  (nonvanishing for  $x \geq x_{3840000} = X_{N^*}(t_0)$ ): the two ranges overlap on the full window  $W_{3840000}$ , so together they cover all  $x \geq x_*$ . Hypothesis (iii) holds by Claim 6.1, since the curved barrier region is contained in  $R \times [0, t_0]$  (Section 6). Theorem 2.1 then gives

$$\Lambda \leq t_0 + \frac{y_0^2}{2} = \frac{893927}{5000000} = 0.1787854. \quad \square$$

## 8 Computational record, corrections, and review boundary

### 8.1 The seven computational propositions

The controlling record for this paper is repository commit [034a37acce703e128e63e2f36520b0c8af6c1c76](#). The primary candidate evidence is mapped in Table 2; derived reconstructions

are mapped separately in Table 3. “Sealed” means that a file is covered by the repository’s [SHA256SUMS](#) inventory and hash-checked by [verify.sh](#); it does not mean that the quick verifier executes that file. The reconstructions reuse or derive from repository sources and are not clean-room implementations under the repository’s adversarial-review protocol. They provide fresh-run and, in most cases, second-toolchain corroboration only; the sealed original sources, certificates, logs, and verifiers remain authoritative for the candidate.

| Proposition | Candidate assertion        | Primary sealed evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.3         | finite window sweep        | <a href="#">src/lemma_sweep_p235711.c</a> ; <a href="#">certificates/</a> ; <a href="#">verifiers/verify_finite_and_binding.py</a> ; <a href="#">logs/finite_and_binding.log</a> ; <a href="#">scripts/run_full_sweep.sh</a>                                                                                                                                                                                                                                                          |
| 4.9         | Dini-cell inequalities     | <a href="#">verifiers/verify_triangle_y_dini_arb.c</a> ; <a href="#">verifiers/verify_triangle_y_dini_logs.py</a> ; <a href="#">logs/triangle_y_dini_180.log</a> ; <a href="#">logs/triangle_y_dini_256.log</a> ; <a href="#">scripts/run_interval_verifiers.sh</a>                                                                                                                                                                                                                   |
| 4.10        | finite-region error        | <a href="#">verifiers/verify_finite_and_binding.py</a> ; <a href="#">logs/finite_and_binding.log</a> (coarse assembly gate); the sealed same-backend file in Table 3 gates the sharper displayed digits                                                                                                                                                                                                                                                                               |
| 5.10        | tail cutoff inequalities   | <a href="#">verifiers/verify_tail_arb.c</a> ; <a href="#">verifiers/verify_tail_arb_logs.py</a> ; <a href="#">logs/tail_arb_256.log</a> ; <a href="#">logs/tail_arb_512.log</a> ; <a href="#">scripts/run_tail_arb.sh</a>                                                                                                                                                                                                                                                             |
| 6.2         | barrier-box error          | <a href="#">barrier/src/verify_uniform_error_01787854.c</a> ; <a href="#">barrier/certificates/uniform_error_256.log</a> ; <a href="#">verifiers/verify_barrier_binding.py</a> ; <a href="#">scripts/run_barrier_replay.sh</a>                                                                                                                                                                                                                                                        |
| 6.5         | tail-exponent inequality   | <a href="#">barrier/src/TloopSinglemat_closed_cert.c</a> ; <a href="#">barrier/certificates/barrier_target_closed.log</a> ; <a href="#">verifiers/verify_barrier_binding.py</a> ; <a href="#">scripts/run_barrier_replay.sh</a>                                                                                                                                                                                                                                                       |
| 6.12        | closed prism decomposition | <a href="#">barrier/src/TloopSinglemat_closed_cert.c</a> ; <a href="#">barrier/src/StoredSumSinglemat_interval.c</a> ; <a href="#">barrier/src/StoredSumTaylorTail_cert.c</a> ; <a href="#">barrier/certificates/barrier_target_closed.log</a> ; <a href="#">barrier/certificates/storedsum_taylor_tail.log</a> ; <a href="#">barrier/certificates/storedsum_provenance.log</a> ; <a href="#">verifiers/verify_barrier_binding.py</a> ; <a href="#">scripts/run_barrier_replay.sh</a> |

Table 2: Primary sealed evidence for the seven candidate assertions at commit [034a37acce703e128e63e2f36520b0c8af6c1c76](#).

| Proposition | Derived standalone reconstruction                                                                       | Seal/execution status                                  |
|-------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 4.3         | <a href="#">independent/prop43/prop43_proof.c</a> and <a href="#">independent/prop43/INVOCATIONS.md</a> | source/schedule sealed; not quick-executed             |
| 4.9         | <a href="#">dan-reworking/code/prop49/</a>                                                              | unsealed                                               |
| 4.10        | <a href="#">independent/prop410/prop410_proof.py</a>                                                    | sealed; quick-executed; same Python backend as primary |
| 5.10        | <a href="#">dan-reworking/code/prop510/</a>                                                             | unsealed                                               |
| 6.2         | <a href="#">independent/prop62/prop62_proof.c</a>                                                       | sealed; quick-executed                                 |
| 6.5         | <a href="#">dan-reworking/code/prop65/</a>                                                              | unsealed                                               |
| 6.12        | <a href="#">dan-reworking/code/prop612/</a>                                                             | unsealed                                               |

Table 3: Derived reconstruction programs. These programs corroborate but do not constitute the primary evidence in Table 2.

The derived reconstruction programs share the following intended discipline:

- all inputs are exact integers and rationals fixed either in the source code or in a complete recorded invocation schedule, and no stored certificate data is read at run time;
- the six C programs use rigorous FLINT/Arb ball arithmetic, with directed endpoint extraction wherever a one-sided bound is formed. Proposition 4.10 is the exception:

both its primary verifier and promoted checker use `mpmath.iv` 1.4.1 under Python 3.12.3 in the canonical container. Because the upstream library labels that interval module experimental and does not guarantee every function, correctness of its transcendental enclosures is an unresolved part of the candidate trust base;

- every acceptance condition is a strict directed comparison (a strict ball inequality, or a strict inequality between exact rationals) and fails closed: an overlapping or indeterminate comparison aborts the run;
- fixed sharp thresholds are compared explicitly against exact rationals where the reconstruction exposes such a gate. The Proposition 4.3 reconstruction is the exception: it prints computed floors and a positivity verdict rather than hard-gating every table entry, which is one reason the sealed canonical certificate remains the authority for Table 1.

The Proposition 4.3 derived reconstruction is the expensive component: its recorded schedule is in [independent/prop43/INVOCATIONS.md](#) and costs roughly three single-core hours before sharding. The remaining reconstruction programs finish in seconds to minutes. Runtime is not a mathematical premise; the exact schedule, source identity, strict exit conditions, and transcript validation are.

The thinnest quantitative margins in the proof are the finite-range post-error margin ( $5.6 \times 10^{-7}$ , Proposition 4.11), the height-transfer ratio ( $1 - \text{ratio} \approx 1.4 \times 10^{-6}$ , Proposition 4.9), and the tail contraction ( $1 - D \approx 2.8 \times 10^{-4}$ , Proposition 5.10). The trust base otherwise has the standard form for computer-assisted proofs of this kind: the cited analytic results, the handwritten reductions, the proof-to-code correspondence, FLINT/Arb and the compiler, and the hardware on which the directed computations run. In addition, this snapshot still assumes valid enclosure behavior for the particular `mpmath.iv` operations in Proposition 4.10; no different transcendental backend checks that proposition. Repetition across derived code paths reduces some implementation and document-drift risks but does not by itself prove either the handwritten correspondence or that backend assumption.

## 8.2 Repository checks at the audited snapshot

The stored quick chain begins at [verify.sh](#); its relevant final assembly and promoted-reconstruction stages are [verifiers/verify\\_assembly\\_1787854.py](#) and [verifiers/verify\\_independent\\_crosscheck.py](#). At commit [034a37acce703e128e63e2f36520b0c8af6c1c76](#), the repository records the following completed checks: the exact three-hypothesis assembly; all 3,149,013 finite rows; the standalone tail verifier at two precisions; all 883 closed barrier prisms; the 7,688 coefficient-component regeneration; exact finite/tail overlap; three promoted derived reconstruction sources (of which the quick verifier executes two); and a separate six-case mutation helper, [verifiers/verify\\_mutation\\_resistance.py](#), that deliberately perturbs constants, documents, and a required program and rejects every tested mutation. This helper is not called by [verify.sh](#) and is not an exhaustive adversarial test of every load-bearing branch. The repository’s status line nonetheless remains

*unreviewed computer-assisted unconditional proof candidate; not an established theorem.*

Here “unconditional” describes the logical form—no conjecture is assumed—and does not assert that the candidate proof is correct.

### 8.3 Corrections incorporated after the supplied exposition

The 22-page July exposition is not the controlling manuscript. The present edition incorporates the later repository corrections most relevant to a referee:

- (1) the computer-assisted components are counted correctly as seven, and each has a complete proof block and a concrete source path;
- (2) the verified-height input is quoted at the exact 3,000,175,332,800 endpoint, and the finite error route deliberately derives the conservative constant 10.50 from Polymath Proposition 6.6(vi) rather than using the incompatible displayed 10.44 route;
- (3) the finite total error is certified directly rather than inferred from rounded component displays; the tail proof uses the directly certified post-error margin, states the endpoint-cap admissibility conditions, and fixes the direction of the comparison with  $1 - D$ ;
- (4) the Dini height-transfer inequality is now a load-bearing assembly gate, with worst certified ratio at most 0.99999860767275095;
- (5) the sharp finite and barrier error constants are separately recomputed, and the sealed finite floors are distinguished from the slightly tighter unsealed reconstruction outputs;
- (6) nonvanishing of  $B_t$  is treated structurally, and the  $t = 0$  endpoint continuity logic and the abstract equivalence  $H = 0 \iff H/B = 0$  when  $B \neq 0$  have partial, **sorry-free** Lean formalizations in [lean/BarrierAnalyticSkeleton.lean](#) and [lean/HypothesisDischarge.lean](#), with scope recorded in [lean/README.md](#). These do not formalize the concrete  $H_t$  constituents or the argument-principle zero-count step;
- (7) the winding increment is written in the clockwise orientation actually used by the sealed code, and the fourth argument of `prop612_proof` is identified as output verbosity only, giving the complete invocation `prop612_proof 0 0.16125 0.1809 1`;
- (8) the complete invocation schedule for Proposition 4.3 was recovered from the sealed sweep runner; the authoritative object is that schedule, not an incidental historical sharding, and the canonical runtime is reported as roughly three single-core hours rather than “under one hour”; and
- (9) mutation resistance and clean finite-regeneration workflows were added, while exploratory lower-time work remained explicitly outside the sealed 0.1787854 claim.

### 8.4 What remains before theorem status

The automated record is unusually extensive, but the repository has not yet received a completed independent expert report. The remaining review scope includes, in particular:

- an Arb reimplement of Proposition 4.10, or a documented operation-level proof that the pinned `mpmath.iv_sqrt`, `pi`, `log`, `exp`, and power operations enclose their exact values on every interval used;
- the exact weld to Polymath 15, Theorem 1.2, including every boundary and normalization convention;

- the native Triangle-to- $|f_t|$  correspondence, the Dini and window transfer reductions, and the uniform all- $N$  tail argument;
- the closed-barrier derivative bounds, quadrature, winding convention, nonvanishing of  $B_t$ , and extension to  $t = 0$ ;
- coefficient, Taylor-remainder, and error-constant provenance, together with the proof-to-code correspondence on every closed domain; and
- a genuinely independent reproduction that does not reuse the candidate source, partitions, or stored tables.

Promoting the four unsealed programs in Table 3 would close a seal and reproducibility gap, but would not turn these derived programs into clean-room checks. The concrete  $\Phi$  measurability, integrability, continuity, and normalization interfaces also remain handwritten. Finally, the contributor list must be agreed before publication; in particular, Dan Romik’s substantial manuscript and program reworking requires an explicit authorship or acknowledgment decision. Until these steps are completed, the mathematically accurate citation describes this work as a computer-assisted proof candidate.

## 8.5 Code availability, release identity, and provenance

At the date of this version, the GitHub repository is access-controlled and commit [034a37acce703e128e63e2f36520b0c8af6c1c76](#) has not been bound to a public release tag or archival DOI. Every repository file or directory path printed in this paper is hyperlinked to that immutable audited commit rather than to the moving `main` branch. On 19 August 2026, `main` still resolved to the same commit, and every explicit program, runner, verifier, transcript, and report path cited in Sections 4–8 was checked to exist there; following the links requires authorized repository access. The commit and tree hashes identify the reviewed state exactly, but they do not make that state independently obtainable. The existing [SH A256SUMS](#) manifest has 381 entries (Git blob `5ad6b2de0c8a63fcdb67c26ba103f24046aa1fba`); the present manuscript and the four unsealed referee-workspace programs are not bound by that manifest. The repository’s current review terms do not grant a public license for repository-original material, so contributor consent and an explicit release license must also be settled before public distribution. A publication release should therefore include current PDF and source hashes, a current immutable tag and DOI, the source and certificate data, complete transcripts and invocation schedules, `SHA256SUMS`, a `RELEASE_MANIFEST.json`, and the tested OCI image and archive digests (including `REVIEW_IMAGE.txt`) under one persistent identifier. It should also carry the canonical Python 3.12.3 and `mpmath` 1.4.1 pins used by Proposition 4.10, enumerate the interval operations invoked, and either validate their enclosure properties or replace that lane with an Arb implementation. Until that archive exists, access to the computational record is limited to authorized reviewers of <https://github.com/judegomila/dbn-lambda-01787854-candidate-audit/tree/034a37acce703e128e63e2f36520b0c8af6c1c76>.

This project builds on and modifies arguments and computational materials in the Mosaic Intelligence preprint [5], distributed under [CC BY 4.0](#); the present author and reviewers are responsible for the modifications, and no endorsement by Mosaic Intelligence is implied. Portions of the computational code descend from the Polymath 15 codebase and remain subject



to the [GNU Lesser General Public License, version 2.1 or later](#). Any code archive accompanying this paper must retain the corresponding copyright, modification, source, and license notices, including the repository’s [THIRD\\_PARTY.md](#). Bundled reference PDFs and arXiv source archives are not relicensed by the repository and must be permission-checked or omitted from any public archive.

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